

Exploratory Data Analysis (EDA) and Descriptive Analytics



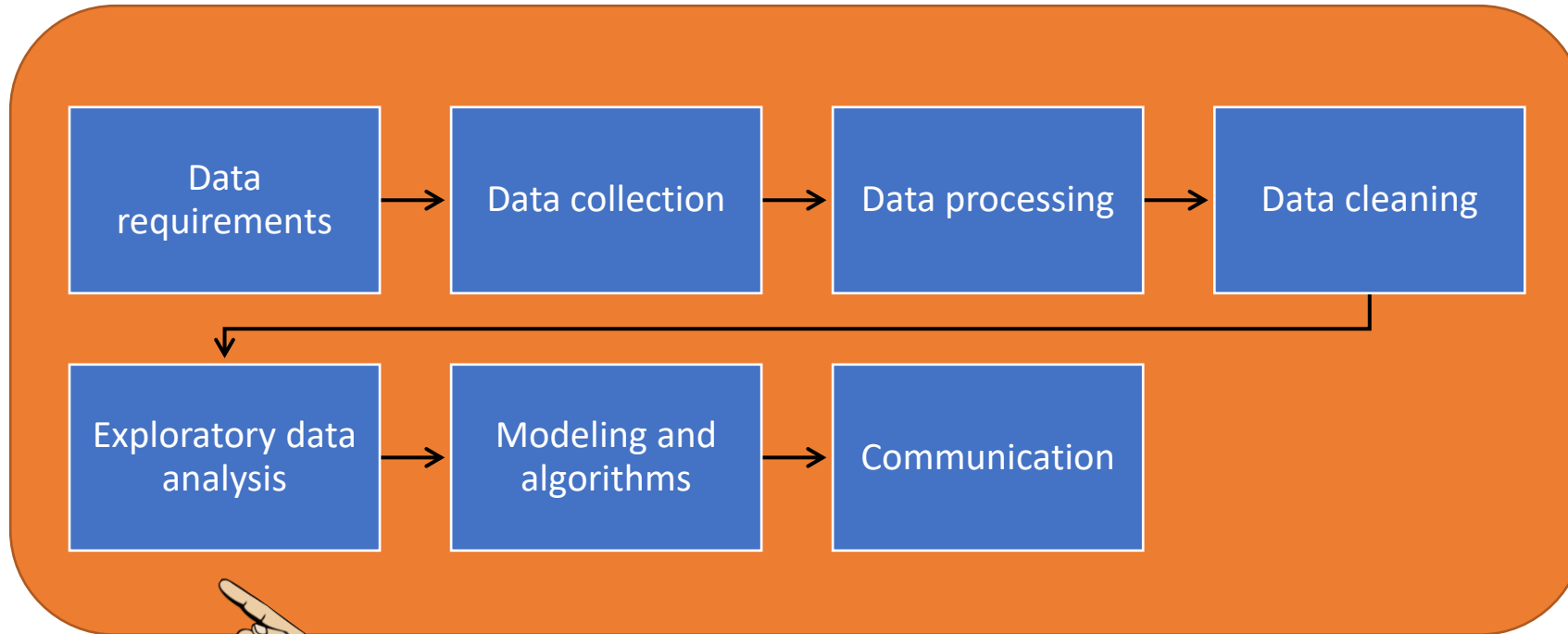
Learning Outcomes for Exploratory Data Analysis and Descriptive Analytics

We will use Descriptive Analytics and show how it helps to perform Exploratory Data Analysis to get a first understanding of our data. We will take a hands-on approach, using Excel to illustrate the concepts we discuss.

You will learn about:

- Exploratory Data Analysis
 - Basic Descriptive Statistics
 - Helpful Graphs and Charts

Process: Data Analysis



Tools

Excel



Zotero



Trello



Methods



Kanban

Considerations



Ethics

Techniques



Descriptive Analytics

Diagnostic Analytics

Predictive Analytics

Prescriptive Analytics

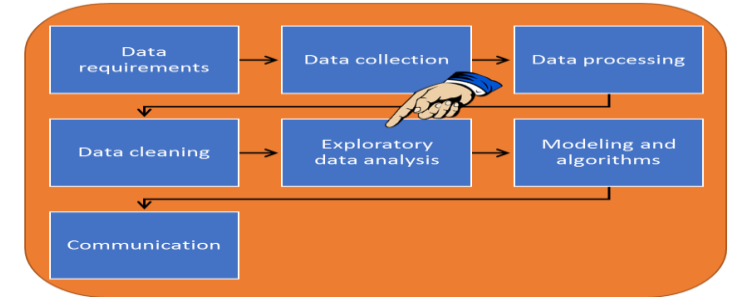
Descriptive Analytics Table of Contents

1. Exploratory Data Analysis using Descriptive Analytics ←
2. Descriptive Statistics
3. Helpful Graphs and Charts

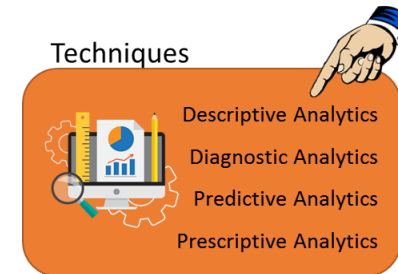
Some considerations

- **Exploratory Data Analysis** is a **step** in the Data Analysis process.
- **Descriptive Analytics** is a **technique** with which to execute this step.
- We can use **different tools** to perform **Descriptive Analytics**.
 - We will learn how to use Excel for this in the second part of the course.

Process: Data Analysis



Techniques

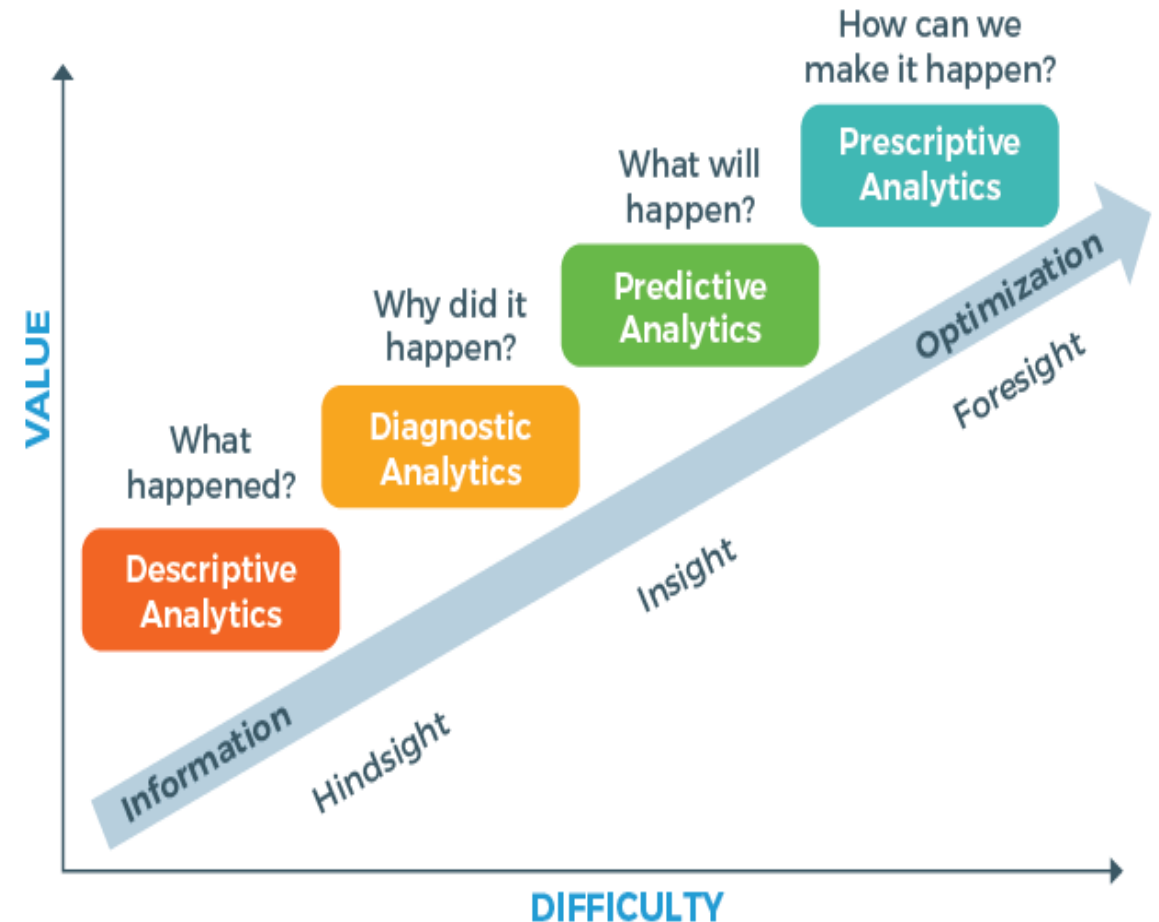


Tools



- We will focus on Descriptive Analytics because
 - It serves as an introduction to Data Analysis
 - It is the easiest one.
 - It helps with Exploratory Data Analysis.
- You will learn more about this and other types of Data Analytics in more depth further along your studies at CUNEF.

Analytic Value Escalator



Exploratory Data Analysis

- Exploratory data analysis (EDA) is the process of **exploring data and investigating its structure to discover patterns and spot anomalies.**
- EDA would then involve **summarizing the data** with the use of **statistics** and **visualization** methods to spot **non-numerical patterns.**



Exploratory Data Analysis

- Ideally, EDA should bring out **insights and realizations** from data that **cannot be obtained** through **formal modeling** and hypothesis testing.
- When done properly, EDA **can dramatically simplify or advance your data science problem** and may even **solve it!**



The Goals Of The Eda Process (EDA: Exploratory Data Analysis)

1. To **question** the data and determine if there are **problems** inherent in the dataset.
2. To **determine if** the **data** on hand is **sufficient** to answer a particular research question or whether additional feature engineering is required.



The Goals Of The Eda Process (EDA: Exploratory Data Analysis)

3. To develop a **framework** for answering the **research** question.
4. To **refine the questions** and/or research problem based on what you have learned about the data.



The Iterative Nature of EDA

- Unlike what most people think, **EDA is not a ‘formal process’**.
- Anything with the word “exploratory” is **not bound by strict rules** and is therefore non-formal.



The Iterative Nature of EDA

- Rather, EDA is **free-form**.
- It is in this stage that you start to **chase ideas** you may have and see which ones **lead somewhere**.



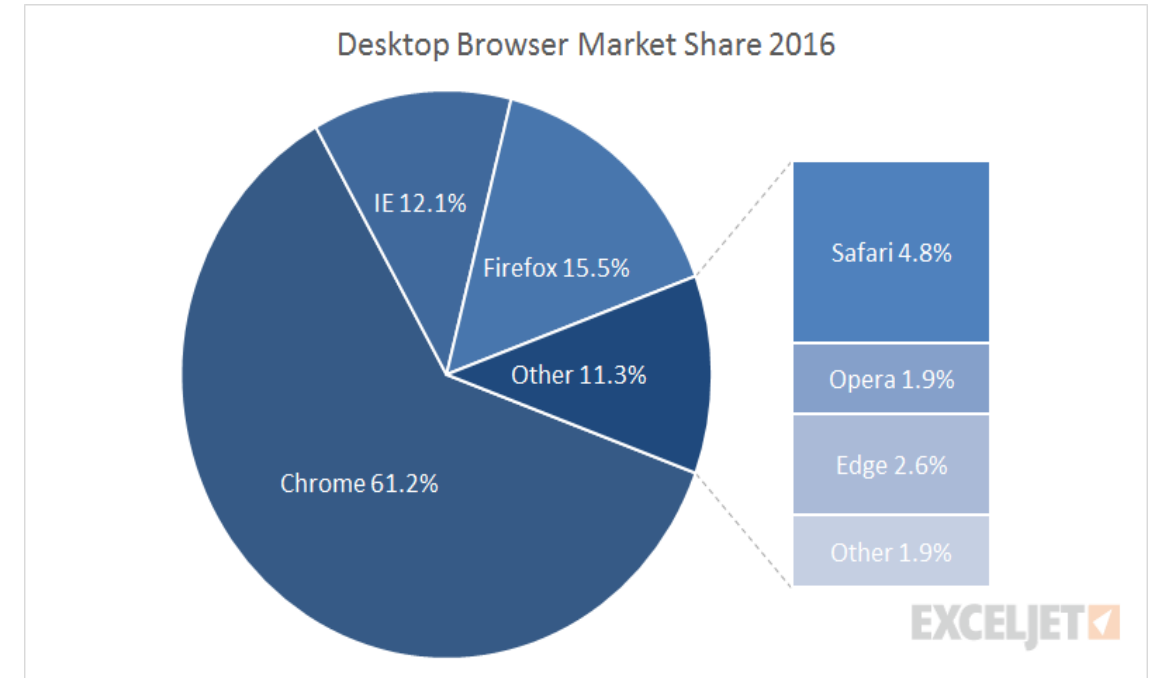
The Iterative Nature of EDA

- Some of these ideas **may not** necessarily **work-out** and you can therefore **start pursuing** or branching from **other viable ideas**.



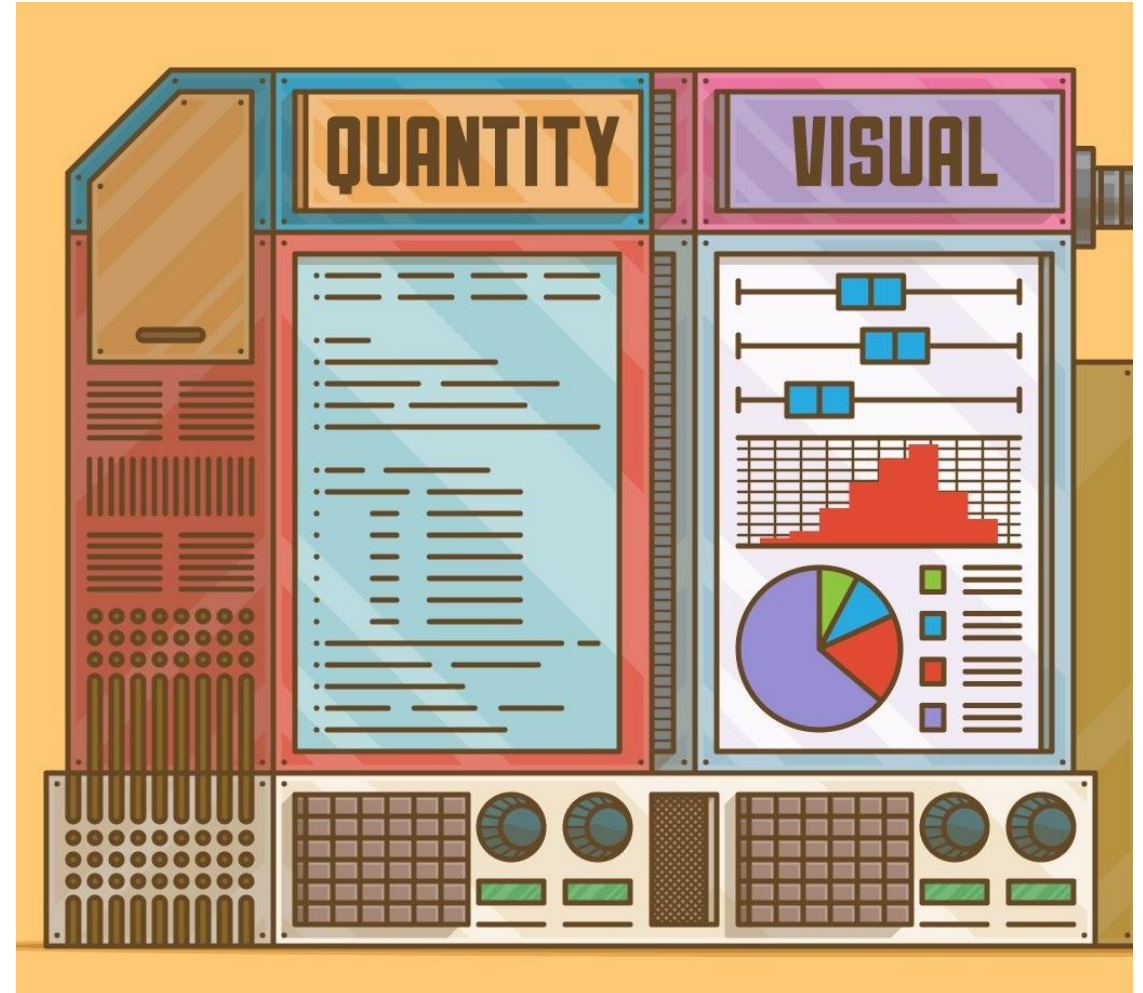
Exploratory Data Analysis

- Exploratory Data Analysis allows us to **understand** the data we are dealing with by **describing and summarizing** the dataset's main characteristics, often through **visual methods** like bar and pie charts, histograms, boxplots, scatterplots, heatmaps, and many more.



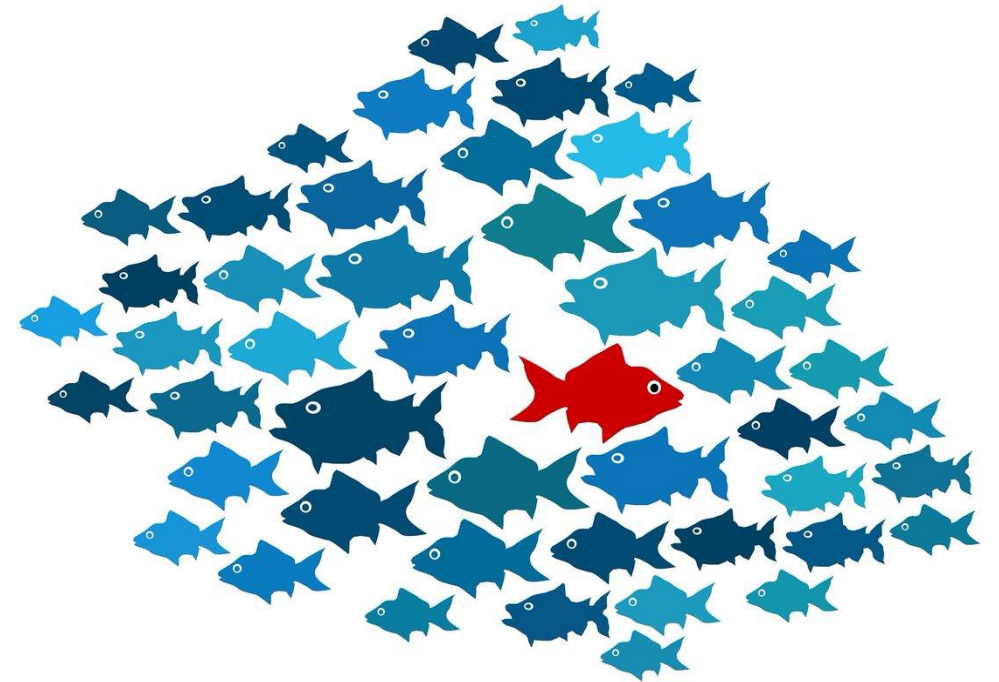
Exploratory Data Analysis

- At a high level, exploratory data analysis (EDA) is the practice of **using visual and quantitative methods to understand and summarize a dataset without making any assumptions about its contents**—a crucial step to take before you dive into machine learning or statistical modeling.



Exploratory Data Analysis will allow us to...

- Maximize **insight** into a dataset (be able to *listen* to your data)
- Uncover underlying **structure/patterns**
- Detect **outliers** and **anomalies**
- Extract and **select important variables**
- **Test** underlying **assumptions** (e.g., business intuition)



How can we get to know our data better?

- Andy Cotgreave, a visual analytics expert at Tableau, recommends to **embrace your inner child** when analyzing data -- ask the data "why," rather than "what".



Exploratory Data Analysis and Statistics

- To be capable of exploring and explain the dataset's features with all its attributes getting insights and efficient numeric summaries of the data, we need to know some statistic concepts

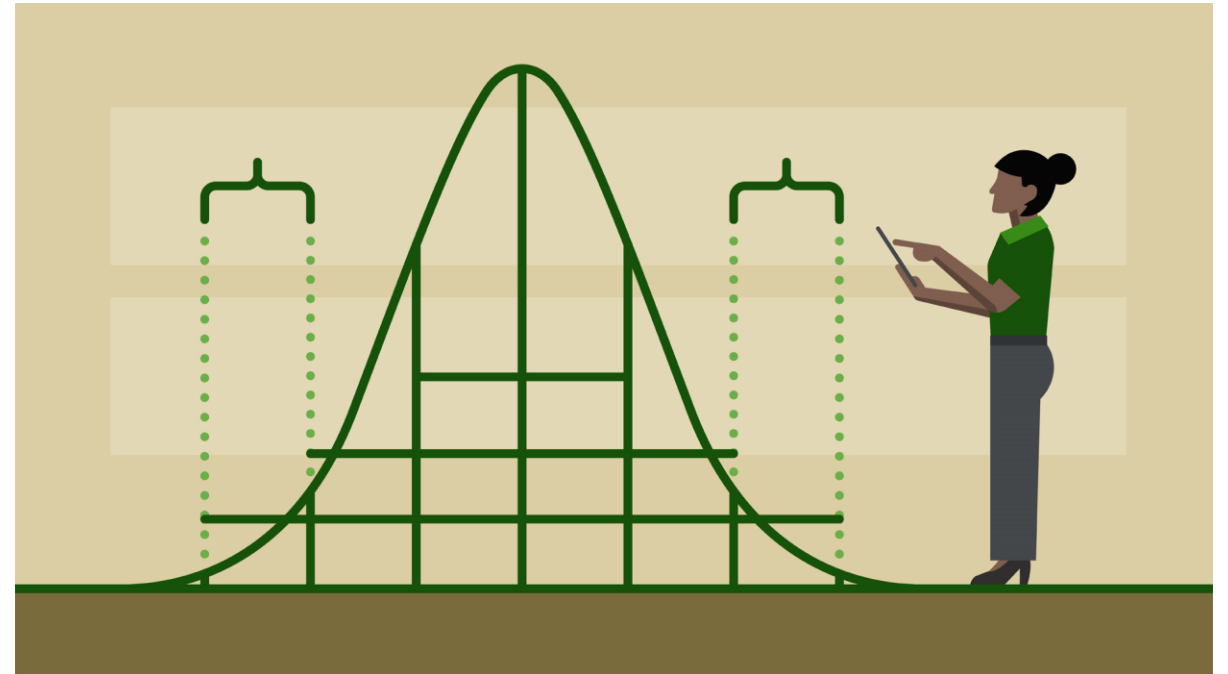
Descriptive Analytics Table of Contents

1. Exploratory Data Analysis using Descriptive Analytics
2. Descriptive Statistics 
3. Helpful Graphs and Charts

Statistics

- **Descriptive statistics:** describe and summarize data
- **Inferential statistics:** use samples to make general conclusions (inferences) about populations.

We will focus on this one



Numerical Summaries of Data (a.k.a. Descriptive Statistics)

- Also named Univariate Analysis (one feature analysis at a time).
- Help **describe** and **understand** the **features** of a specific dataset, by giving **short numeric summaries about the sample and measures of the data.**



Numerical Summaries of Data (a.k.a. Descriptive Statistics)

- Descriptive statistics are mere **exploration** as they do not allow us to make conclusions beyond the data we have analyzed or reach conclusions regarding any hypotheses we might have made.



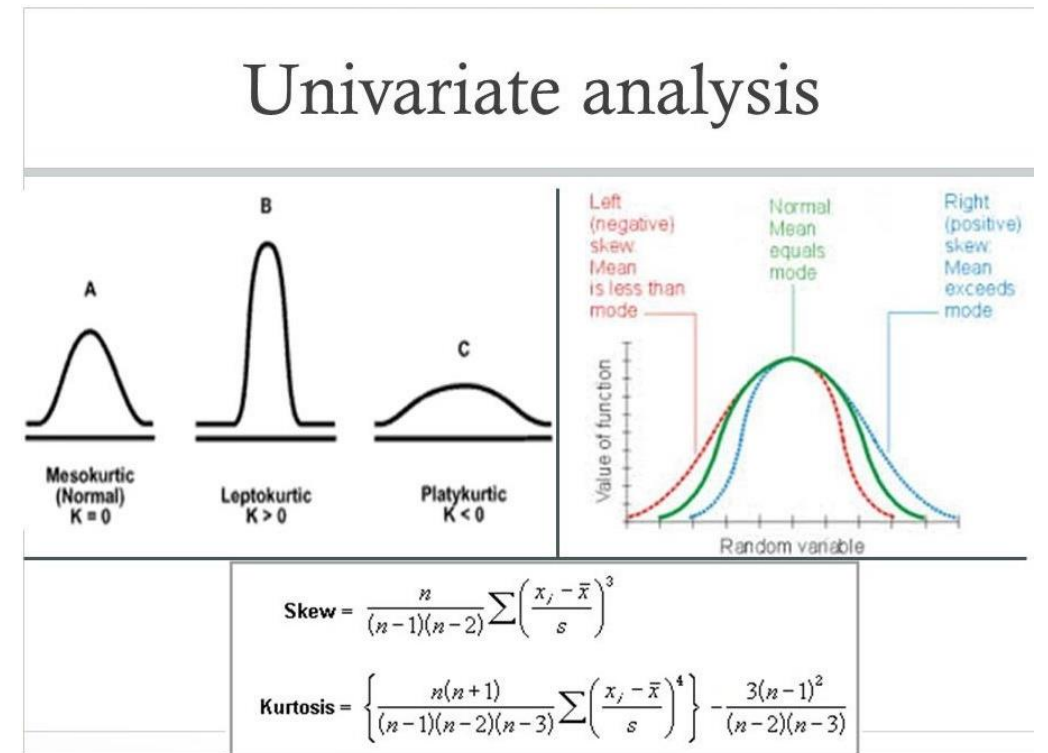
Numerical Summaries of Data (a.k.a. Descriptive Statistics)

- Numerical (Quantitative) and Categorical (Qualitative) variables, as we will see later, have different descriptive statistics approaches.

VARIABLE			
NUMERICAL		CATEGORICAL	
Made of numbers		Made of letters/words	
CONTINUOUS	DISCRETE	ORDINAL	NOMINAL
Temperature	no. of rainy days	Days of week	Blood type
Income	no. of children	Pain scale	Brands
Height	no. of TVs	Ratings	Colors

Numerical Summaries of Data (a.k.a. Descriptive Statistics)

1. Central Tendency measures.
2. Variation or Variability measures.
3. Relative Standing measures.



Numerical Summaries of Data (a.k.a. Descriptive Statistics)

1. Central Tendency measures:

They are computed to give a **center** around which the measurements in the data are distributed.

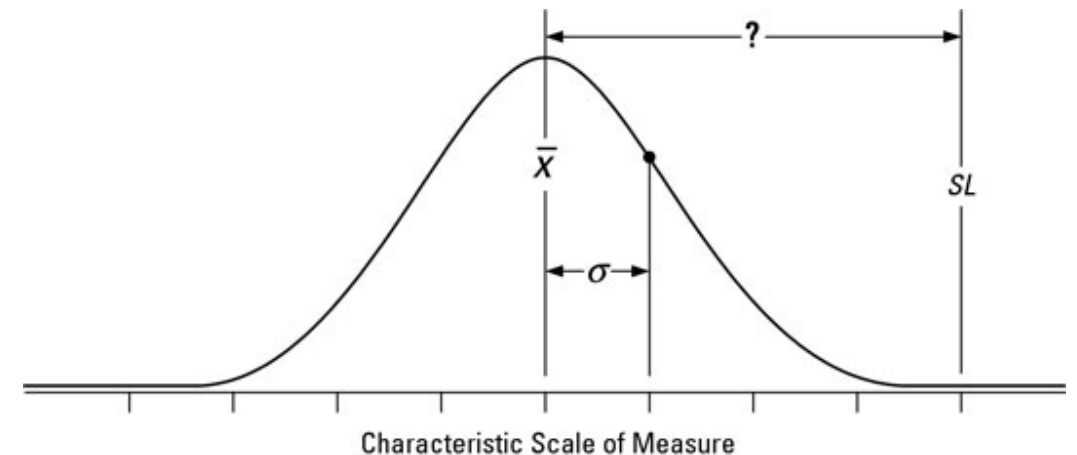
Average Formula

$$\text{Average} = \frac{(a_1 + a_2 + \dots + a_n)}{n}$$



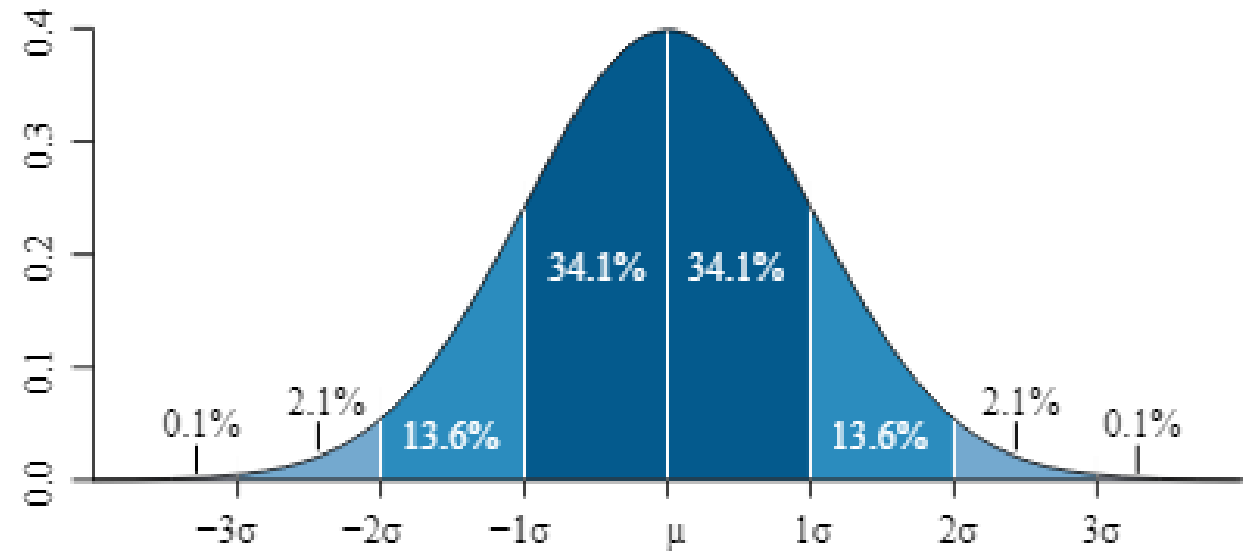
Numerical Summaries of Data (a.k.a. Descriptive Statistics)

2. Variation or Variability measures: They describe **data spread**, or how far away the measurements are from the center.



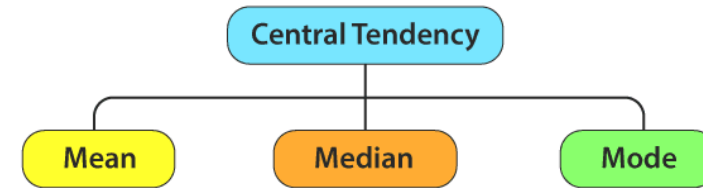
Numerical Summaries of Data (a.k.a. Descriptive Statistics)

3. Relative Standing measures:
They describe the **relative position** of specific measurements in the data.

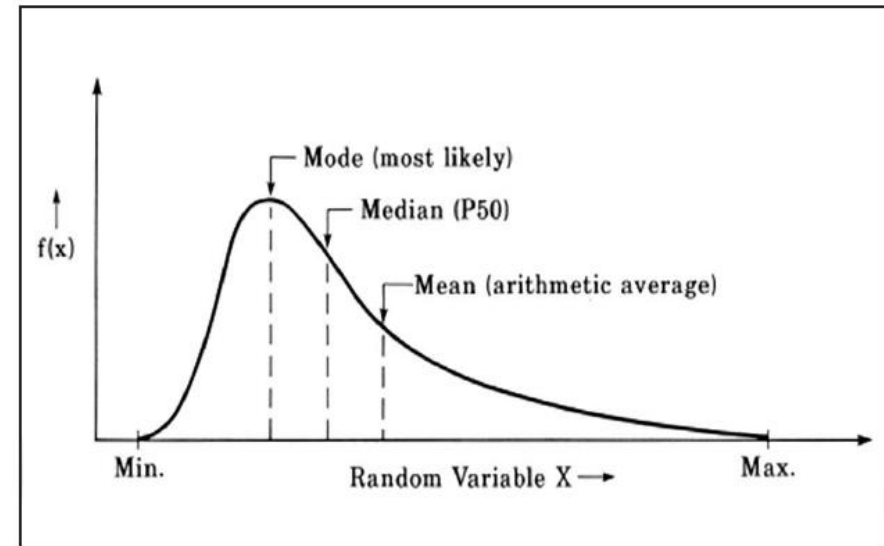


1. Central Tendency measures

CENTRAL TENDENCY



1. Mean (average)
2. Median (center value)
3. Mode (most common)



1. Central Tendency measures - Mean

Mean (average): The total sum of values divided by the total observations. The mean is highly sensitive to the outliers.

$$\text{mean} = \frac{\text{sum of data}}{\# \text{ of data points}}$$

Here's the same formula written more formally:

$$\text{mean} = \frac{\sum x_i}{n}$$

Example: the mean of the values 1, 2, 2, 3, 4, 5 is:

- $1+2+2+3+4+6= 18$
- Total observations= 6
- $18/6=3$
- So, the mean is 3

1. Central Tendency measures

Median (center value):

The middle number;
found by ordering all data
points and picking out the
one in the middle (or if
there are two middle
numbers, taking the mean
of those two numbers).

Example: the median of the values 2, 7, 3, 5, 4, 1, 6 is:

- First, we order them:
1, 2, 3, 4, 5, 6, 7
- Then we find the middle value: there are 7 values, so the middle one is the one in position #4:

1, 2, 3, 4, 5, 6, 7



- The median is 4

1. Central Tendency measures - Mode

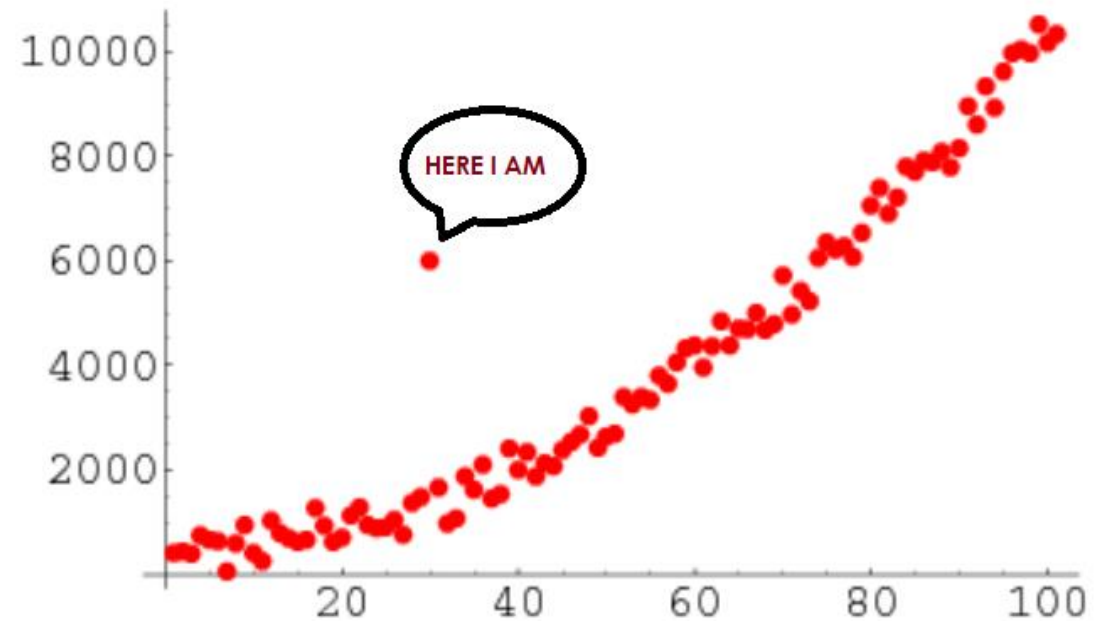
Mode (most common): The values most frequently observed. There can be more than one modal value in the same variable.

Example: the mode of the values 1, 2, 2, 3, 4, 5 is the most frequent number—that is, the number that occurs the highest number of times.

- The Mode is 2

A note on Outliers

- In statistics, an **outlier** is a **data point** that **differs significantly** from other observations.
- May be due to **variability** in the measurement, or it may indicate **experimental error**; the latter are sometimes excluded from the data set.
- An outlier **can cause serious problems** in statistical analyses.



2. Variation or Variability measures

1. Standard deviation
2. Skewness
3. Minimum
4. Maximum



2. Variation or Variability measures

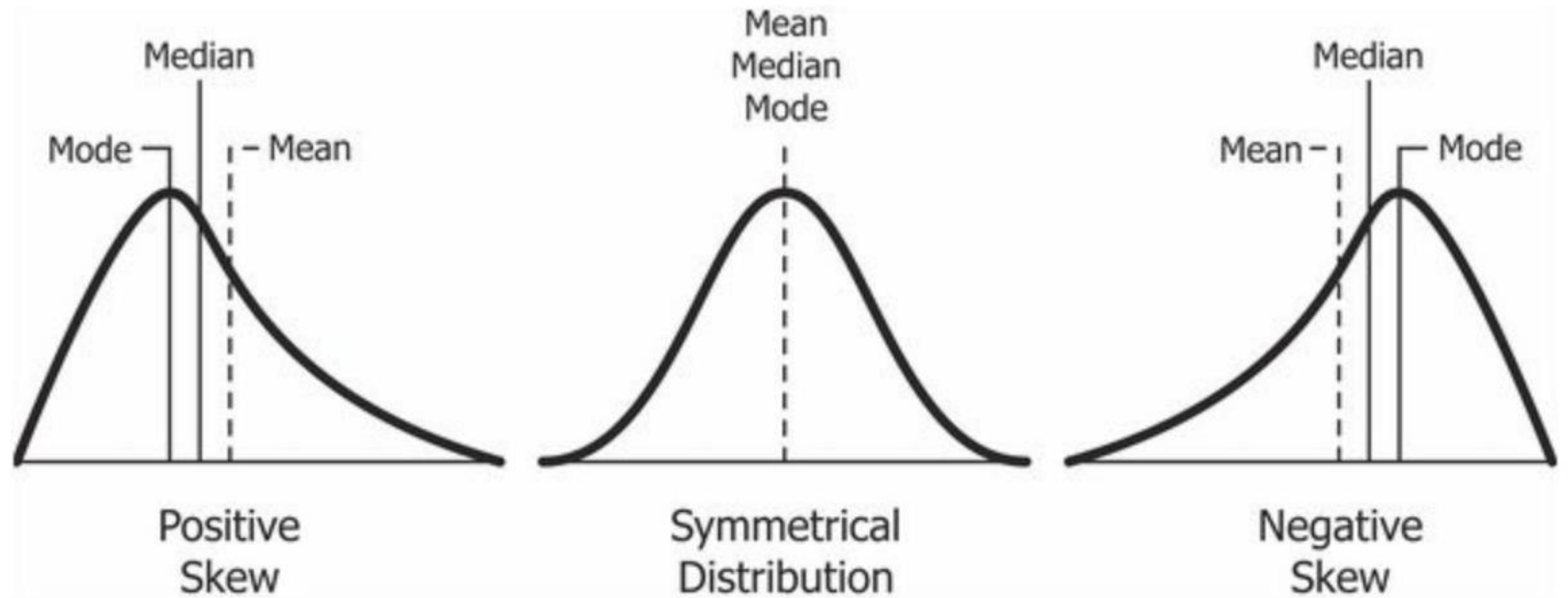
1. **Standard deviation:** a measure of the average distance between the values of the data in the set and the mean.
2. This is a **very useful statistic** because it provides a handy measure of **how much spread there is in the scores in the distribution**.
3. When combined, the mean and the standard deviation provide a pretty good picture of what the distribution of scores is like.

Standard Deviation Formula


$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{N}}$$

2. Variation or Variability measures

2. Skewness:
measures the
symmetry of the
distribution.



2. Variation or Variability measures

3. Minimum: The smallest or lowest value.

4. Maximum: The greatest or highest value.

Example: with the values 2, 7, 3, 5, 4, 1, 6 is:

The Minimum is 1 and the Maximum is 7.

3. Relative Standing measures

- **Percentiles:** measure indicating the value below which a given percentage of observations in a group of observations fall.
- For example, the 20th percentile is the value (or score) below which 20 percent of the observations may be found.

Understanding Score Percentiles

A score percentile represents the percentage of scores that are equal or below a certain score within a given sample.

Example: The 75th percentile SAT score for incoming freshmen is 1400.

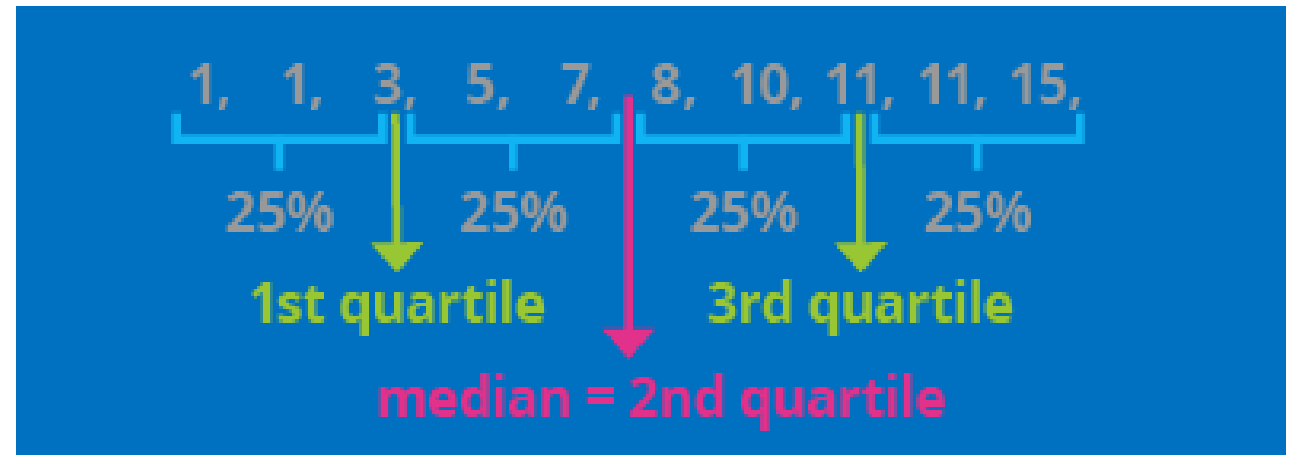


75% of students
Scored 1400 or below

75th percentile
(25% of students)
Scored above 1400

3. Relative Standing measures

- **Quartiles:** The 25th percentile is also known as the first quartile (Q1), the 50th percentile as the median or second quartile (Q2), and the 75th percentile as the third quartile (Q3).



Some potential problems with these measures

- Sometimes, they can give the wrong idea about the data.
- Which is why Exploratory data analysis always incorporates graphs besides numerical measures.
- Sometimes, to get the whole picture you need both!



Let me show you an example

- Let's say we have 4 Datasets (I to IV)
- For each we have 11 measurements of variables X and Y.



Ascombe's quartet



I		II		III		IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

Let's calculate some Descriptive Statistics for each dataset

Property	Values	Accuracy
Mean of X	9	Exact
Mean of Y	7.50	Exact
Sample variance of X	11	Exact
Sample variance of Y	4.125	Exact
Correlation between X and Y	0.816	To 3 decimal places
Linear regression line	$y = 3.00 + 0.500x$	to 2 and 3 decimal places, respectively

So, the datasets are probably very similar, right?

Property	Values	Accuracy
Mean of X	9	Exact
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Sample variance of X	11	Exact
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Correlation between X and Y	0.816	To 3 decimal places
Linear regression line	$y = 3.00 + 0.500x$	to 2 and 3 decimal places, respectively

I		II		III		IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
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13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
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Different data with identical summary statistics
The four datasets are equal when looked through the lens of simple summary statistics, but they significantly differ from each other when graphically plotted

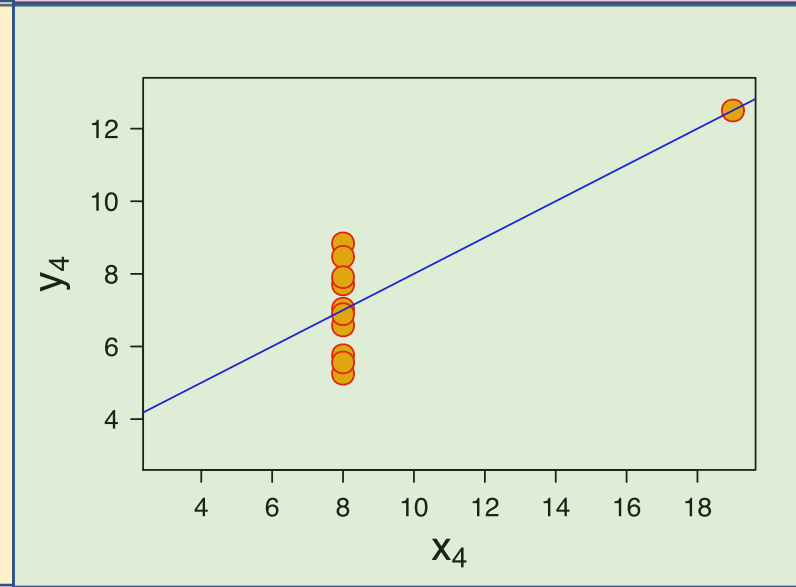
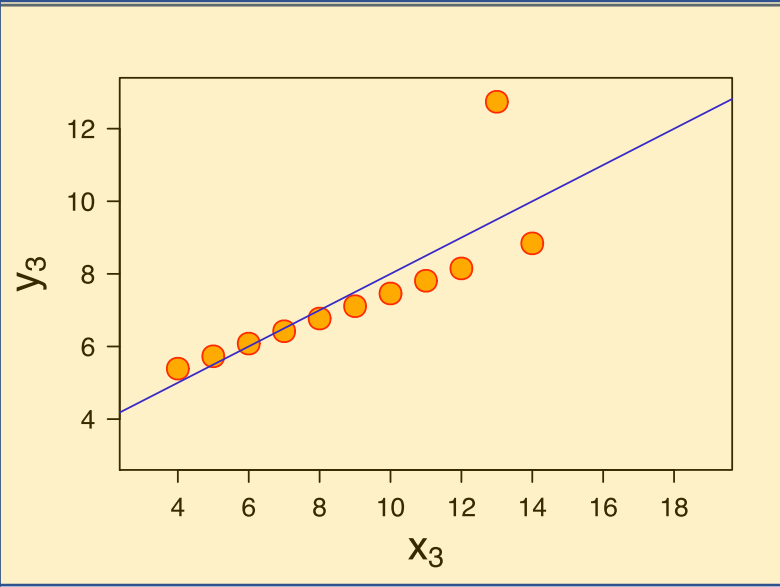
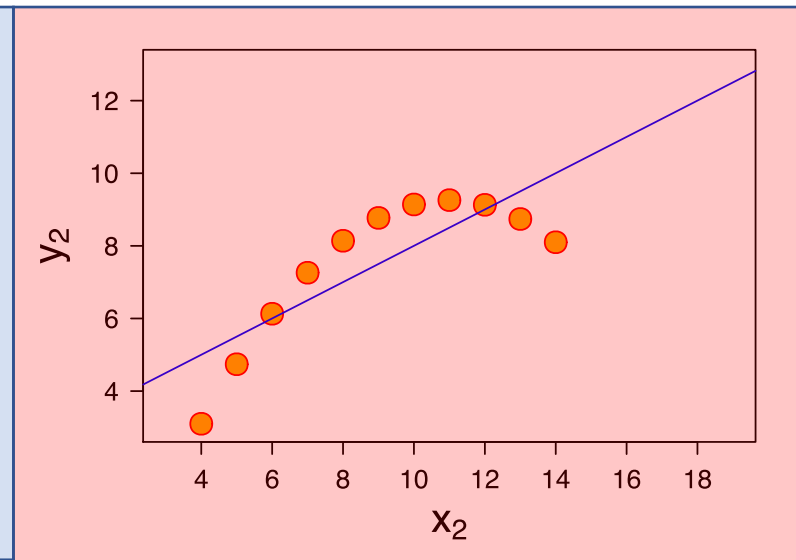
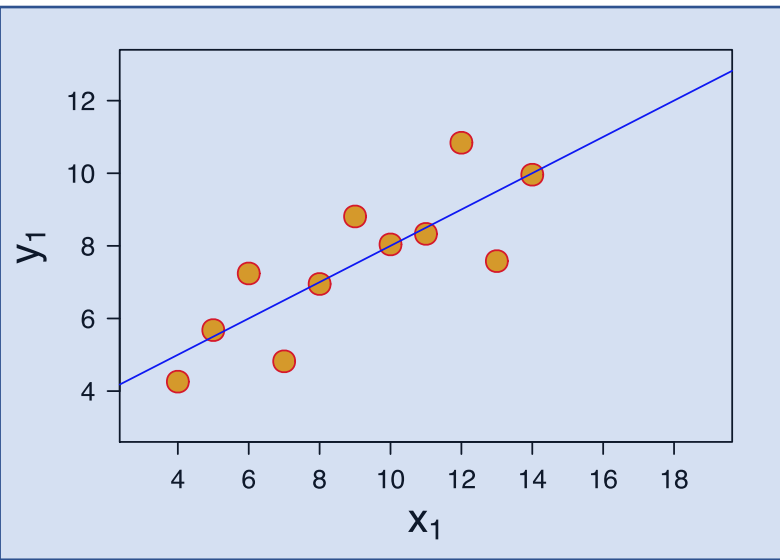
Let's create a graph for each dataset using Excel



Ascombe's quartet

All four sets are identical when examined using simple summary statistics, but vary considerably when graphed

I		II		III		IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
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14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
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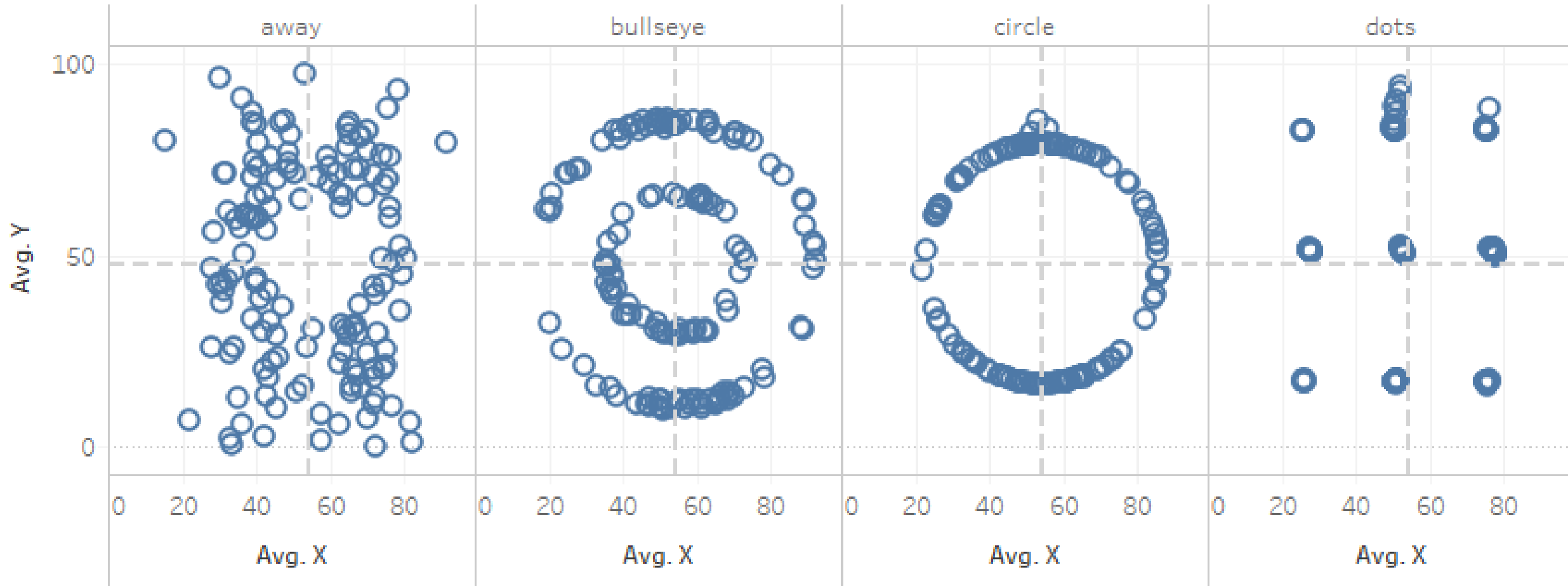


Anscombe's quartet

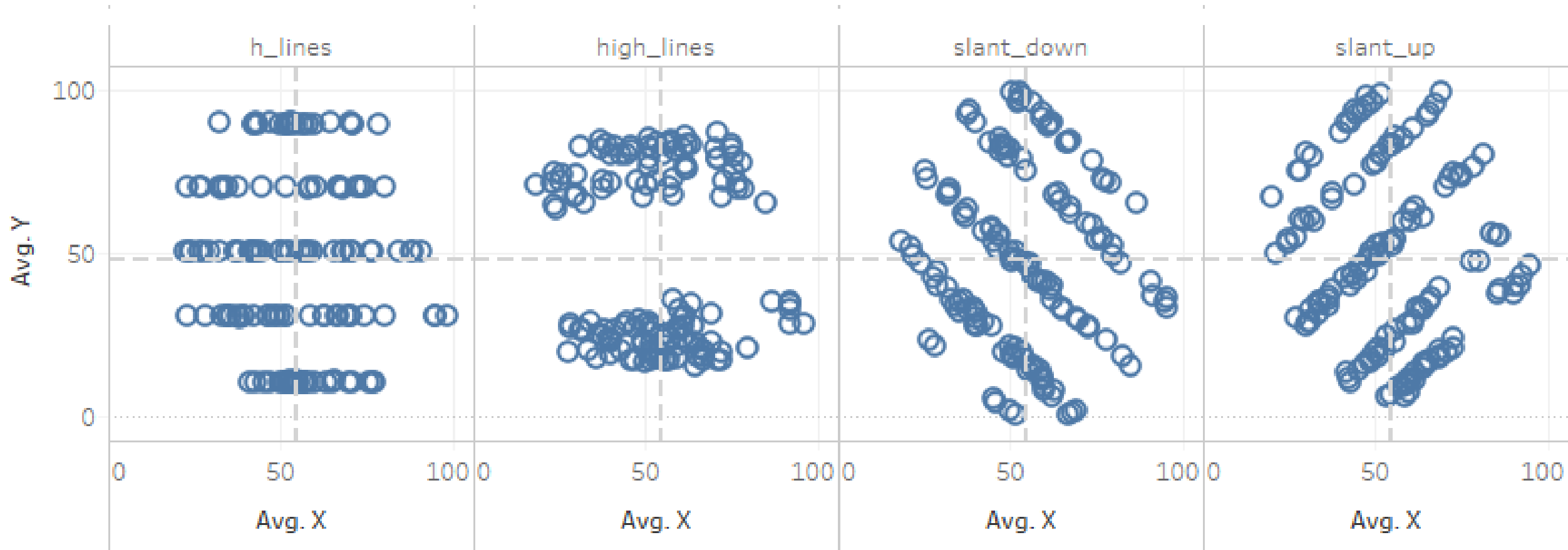
- This example, known as Anscombe's quartet, was formulated in 1973 by the English statistician Francis Anscombe.
- He did it to demonstrate both the **importance of graphing data before analyzing it** and the **effect of outliers** and other influential observations on statistical properties.



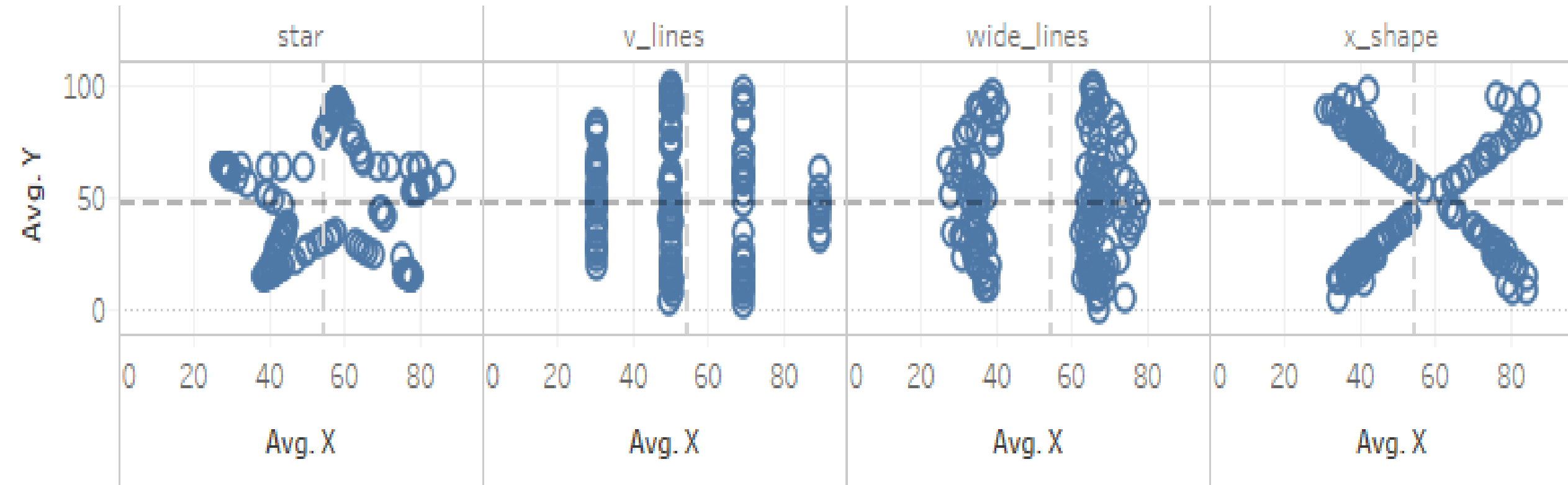
Datasaurus Dozen



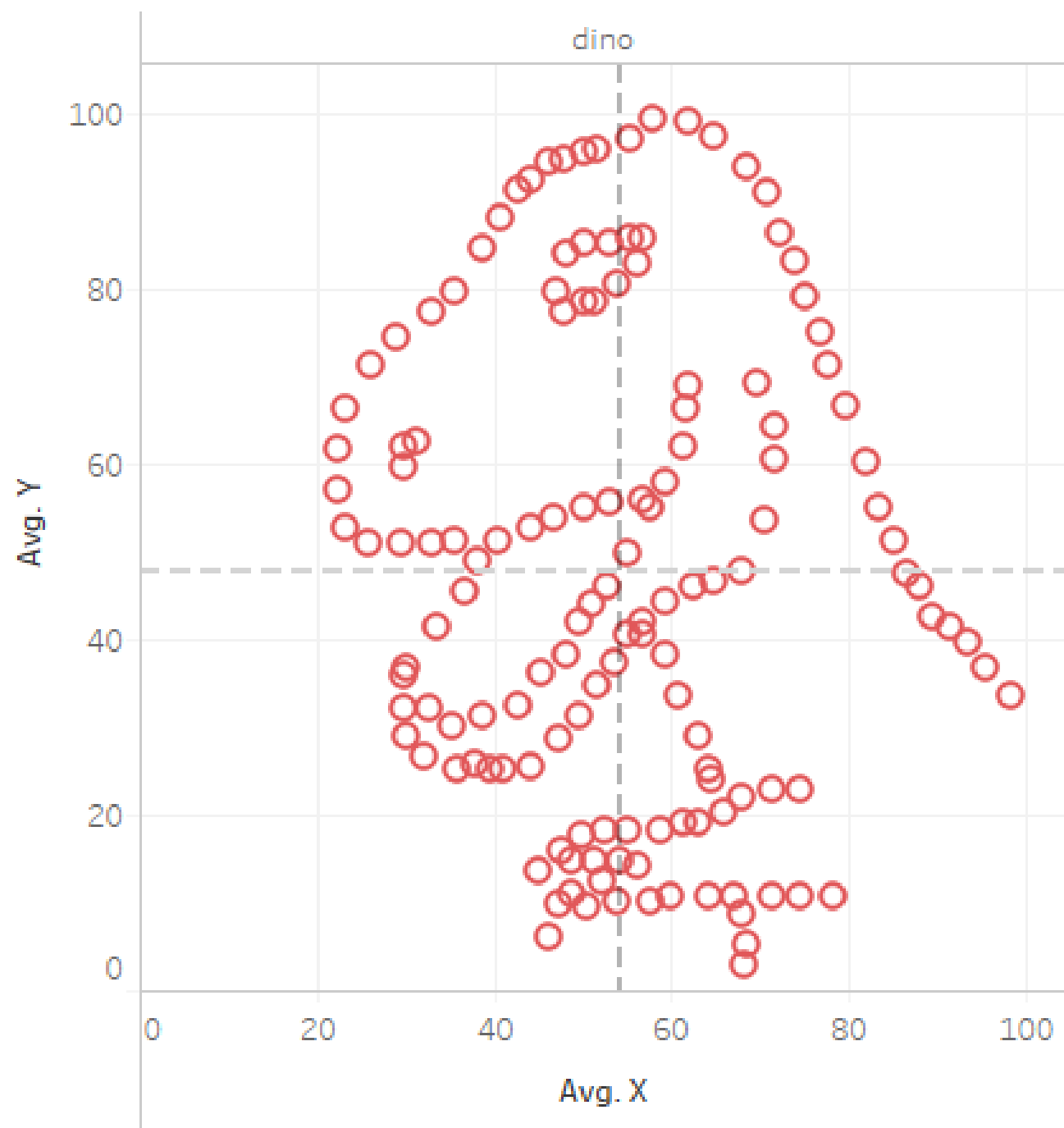
These datasets also have the same descriptive statistics



These datasets also have the same descriptive statistics



This one too!
You can even draw
a dinosaur!



Let's see another example of why it pays to explore your data to get to know it

- Oftentimes, **we are pressed for time and rely on our first impressions.**
- Having some numbers confirming our first impression **can lead to the wrong explanation!**

First Impressions Count!



The example concerns a study of **gender bias** among **graduate school admissions** to University of California, Berkeley.

The **admission figures** for the fall of 1973 **showed** that **men** applying were **more likely** than women **to be admitted**, and the **difference** was so large that it was **unlikely** to be **due to chance**.



Let's take a look at the data

University of Berkley 1973 Admission Stats	Male	Female
Applicants	2691	1835
Admitted	1278	659
Success Rate	47%	36%

- The data looks biased toward men being admitted more often than women.
- If we act on this information alone, we could be taking the wrong course of action!

- When examining the individual departments, it appeared that six out of 85 departments were significantly biased against men, whereas four were significantly biased against women. In total, the pooled and corrected **data showed a small but statistically significant bias in favor of women.**
- The data from the six largest departments are listed below.

	Men		Women	
Department	Applicants	Success Rate	Applicants	Success Rate
I	825	62%	108	82%
II	560	63%	25	68%
III	325	37%	593	34%
IV	417	33%	375	35%
V	191	28%	393	24%
VI	373	60%	341	70%
TOTAL	2691	47%	1835	36%

	Men		Women	
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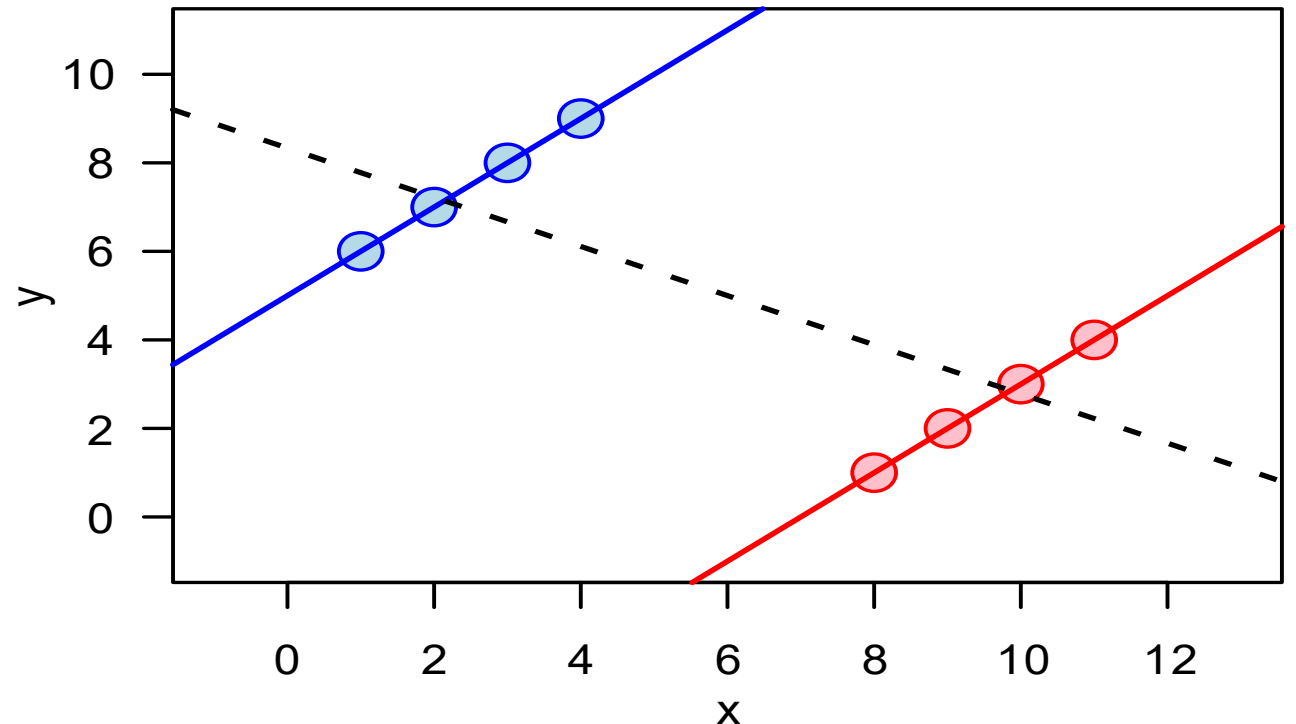
A research paper by Bickel et al. concluded that **women tended to apply to more competitive departments with lower rates of admission**, even among qualified applicants (such as in the English department), **whereas men tended to apply to less competitive departments with higher rates of admission** (such as in the engineering department)

This has been called “The Simpson’s Paradox”

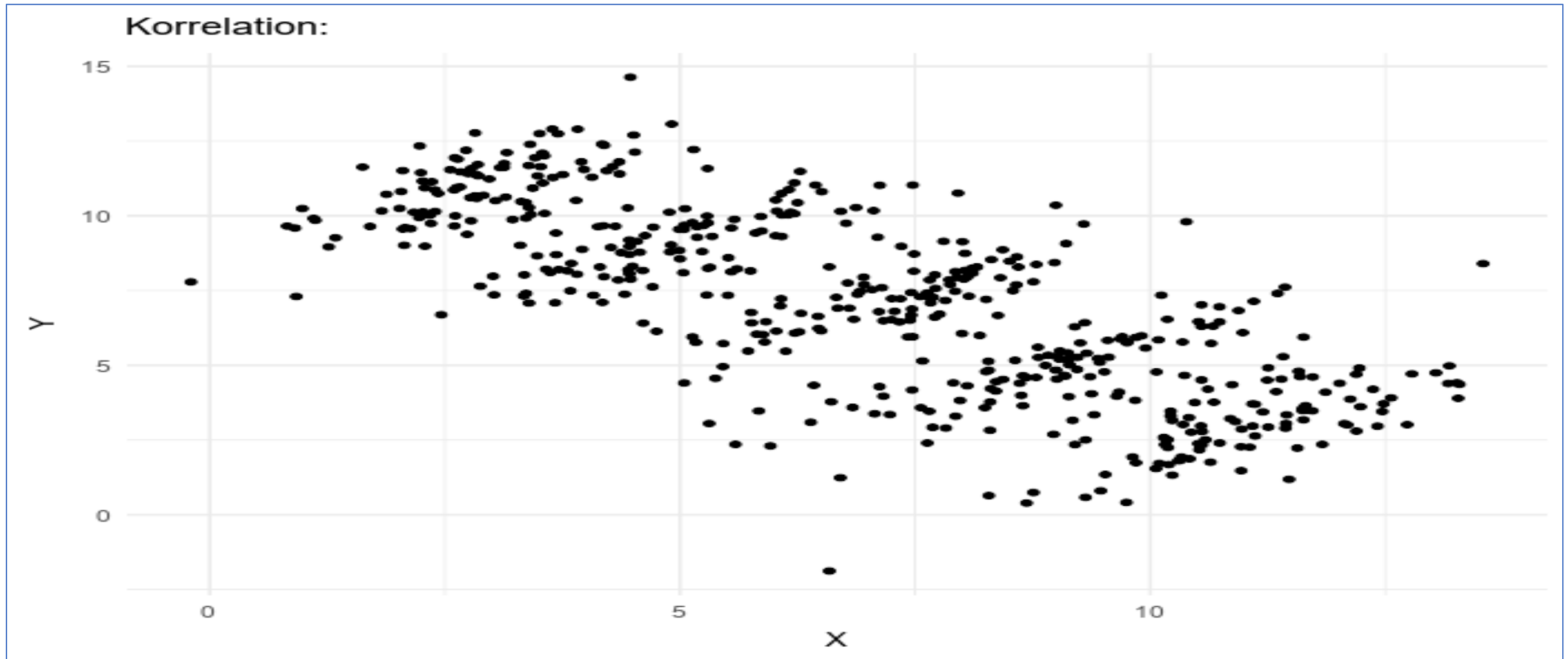


“The Simpson’s Paradox” is a phenomenon in probability and statistics in **which a trend appears in several groups of data but disappears or reverses when the groups are combined.** This result is often encountered in social-science and medical-science statistics

A positive trend appears for two separate groups (**blue** and **red**), whereas a negative trend (**black, dashed**) appears when the groups are combined.



The “The Simpson’s Paradox”



Let's see the Simpson's paradox in a real data set

Download canvas's "penguins - simpson" excel file



penguins - simpson



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1. Exploratory Data Analysis using Descriptive Analytics
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3. Helpful Graphs and Charts 

Helpful Graphs and Charts

1. [Box Plots](#)
2. [Bar Charts](#)
3. [Histograms](#)
4. [Time Series](#)
5. [Scatter plot](#)
6. [Pie Chart](#)
7. [Geospatial](#)



Helpful Graphs and Charts

Types of
charts
Excel can
handle

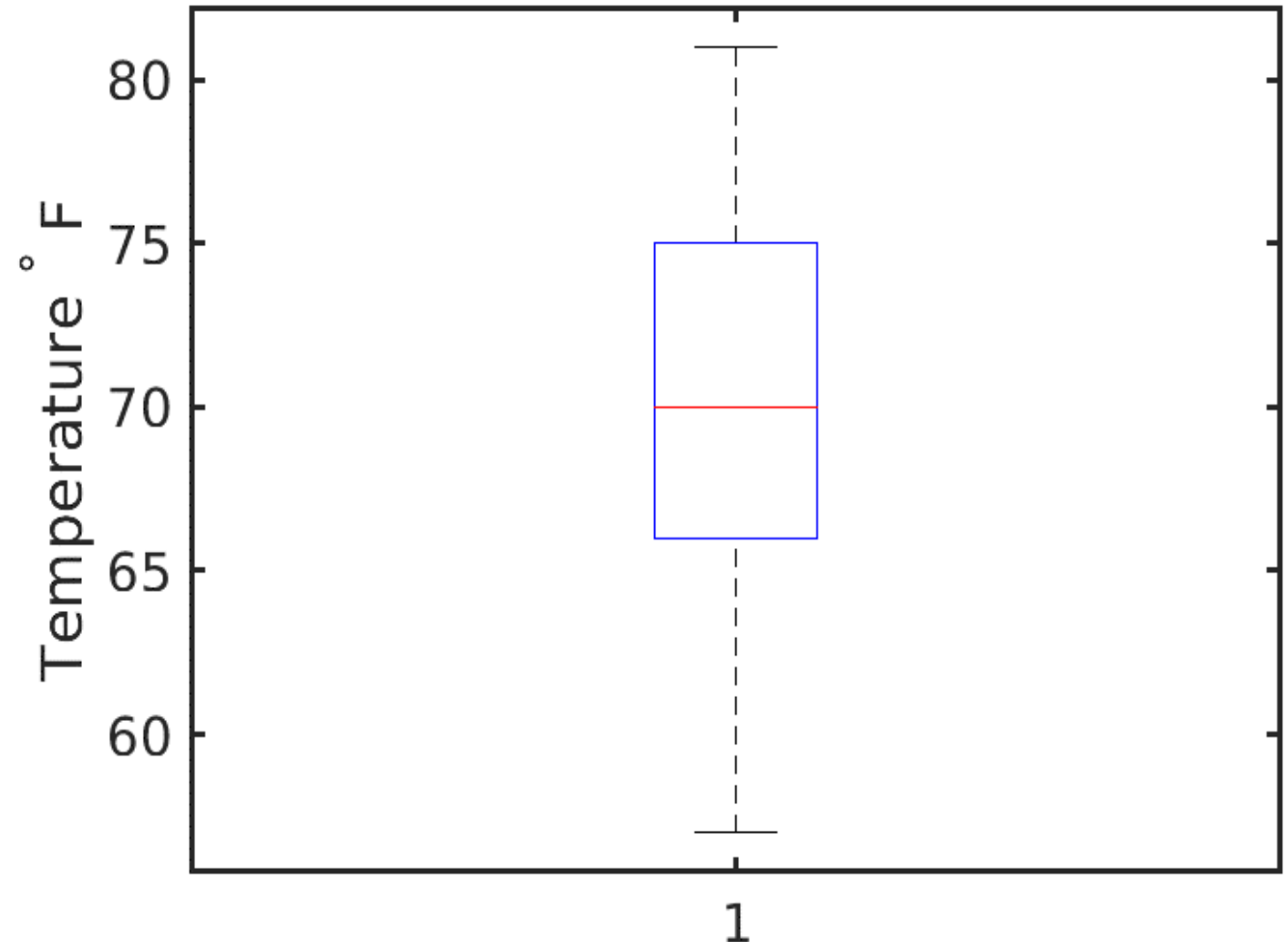


Box Plots

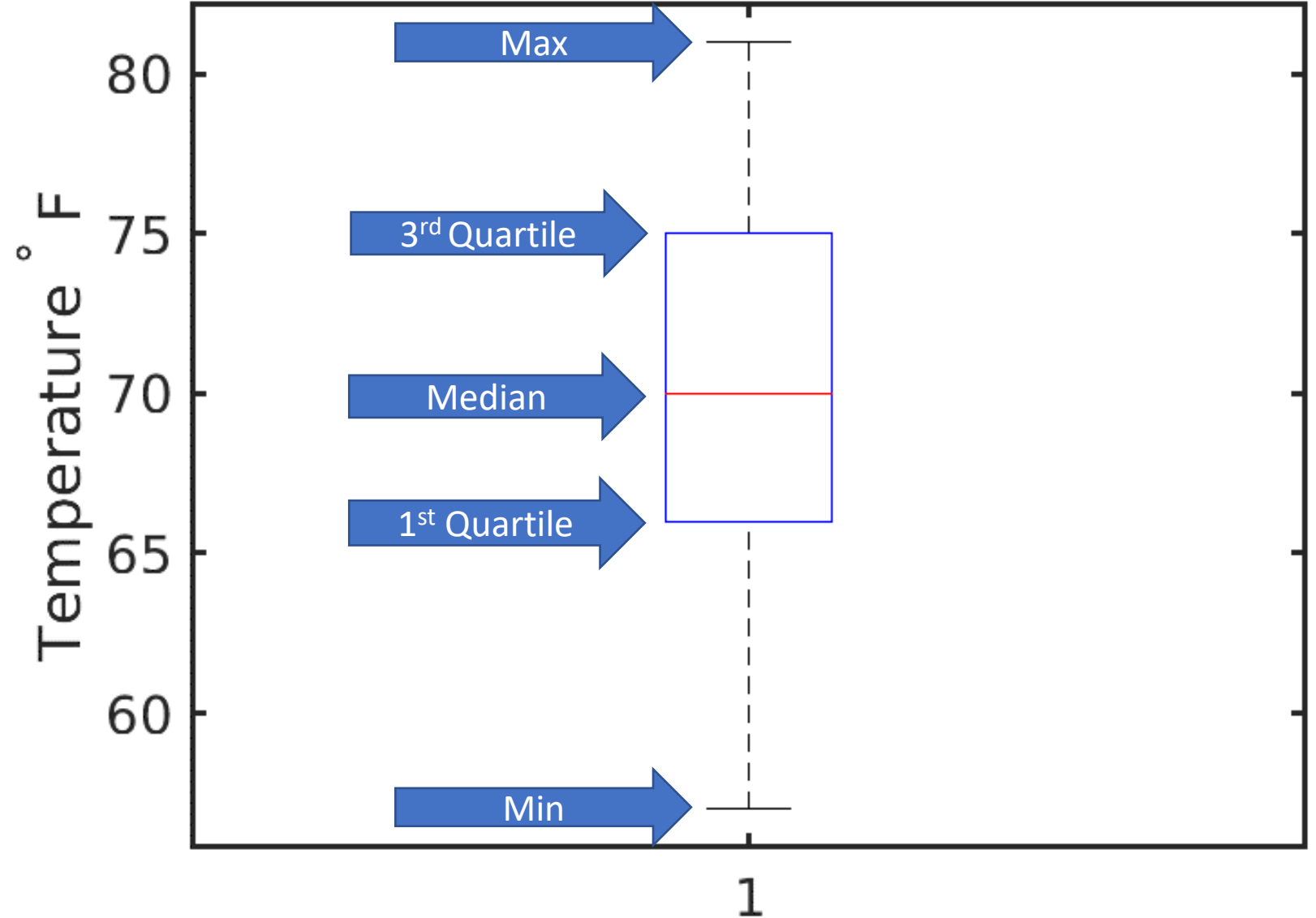
- A box plot or boxplot is a method for **graphically** depicting groups of **numerical data** through their **quartiles**.
- It also depicts the **minimum** and the **maximum** values.



penguins - graphics



Box Plots

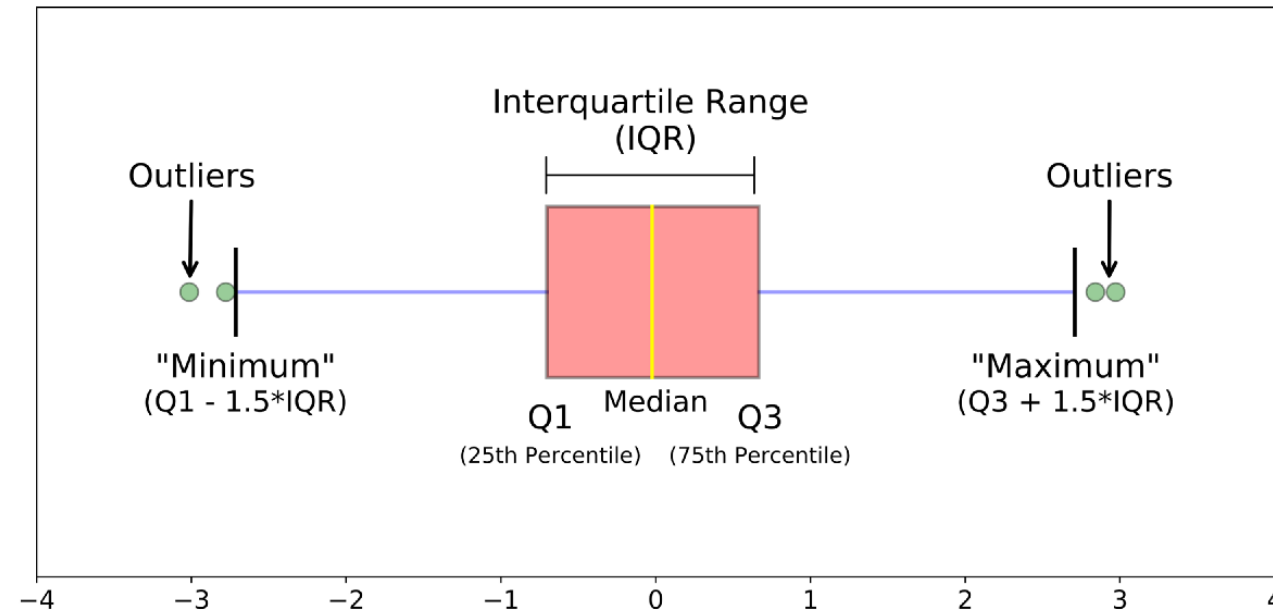


Box Plots – Use

- When you want to **visualize** the **distribution** of **quantitative values** in a field.
- When you want to **compare** across different **categorical variables**.
- When you want to **identify outliers**.



penguins - graphics

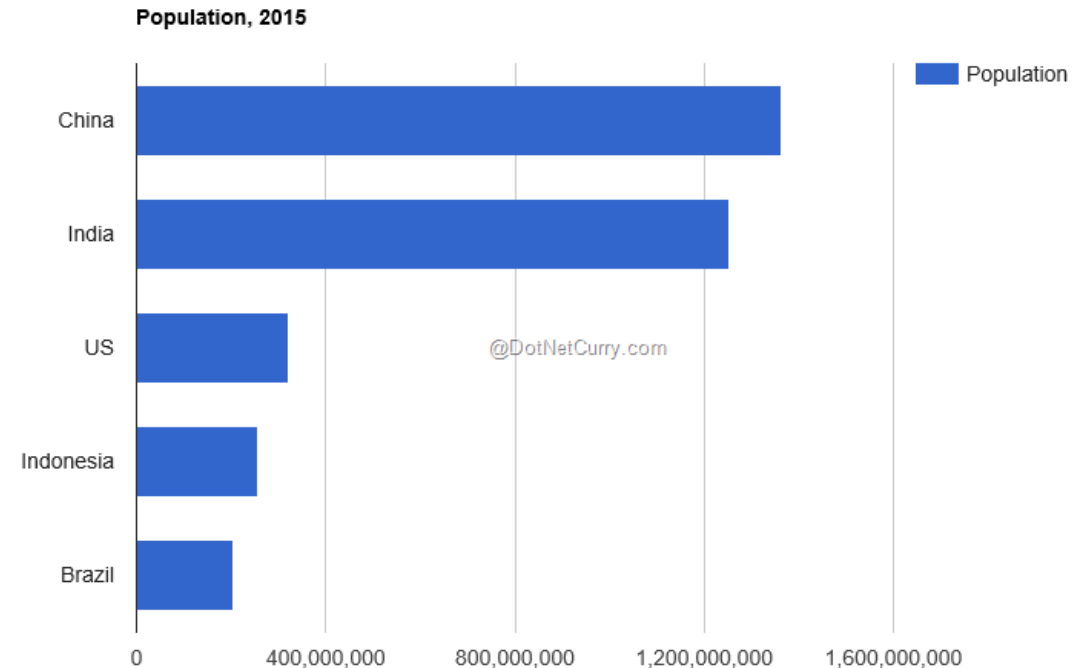


Bar/Column Chart

- A bar chart or bar graph is a chart that **presents categorical data** with rectangular bars with **heights or lengths proportional to the values** that they represent.
- The **bars** can be plotted **vertically or horizontally**.



spain salary by comunidad

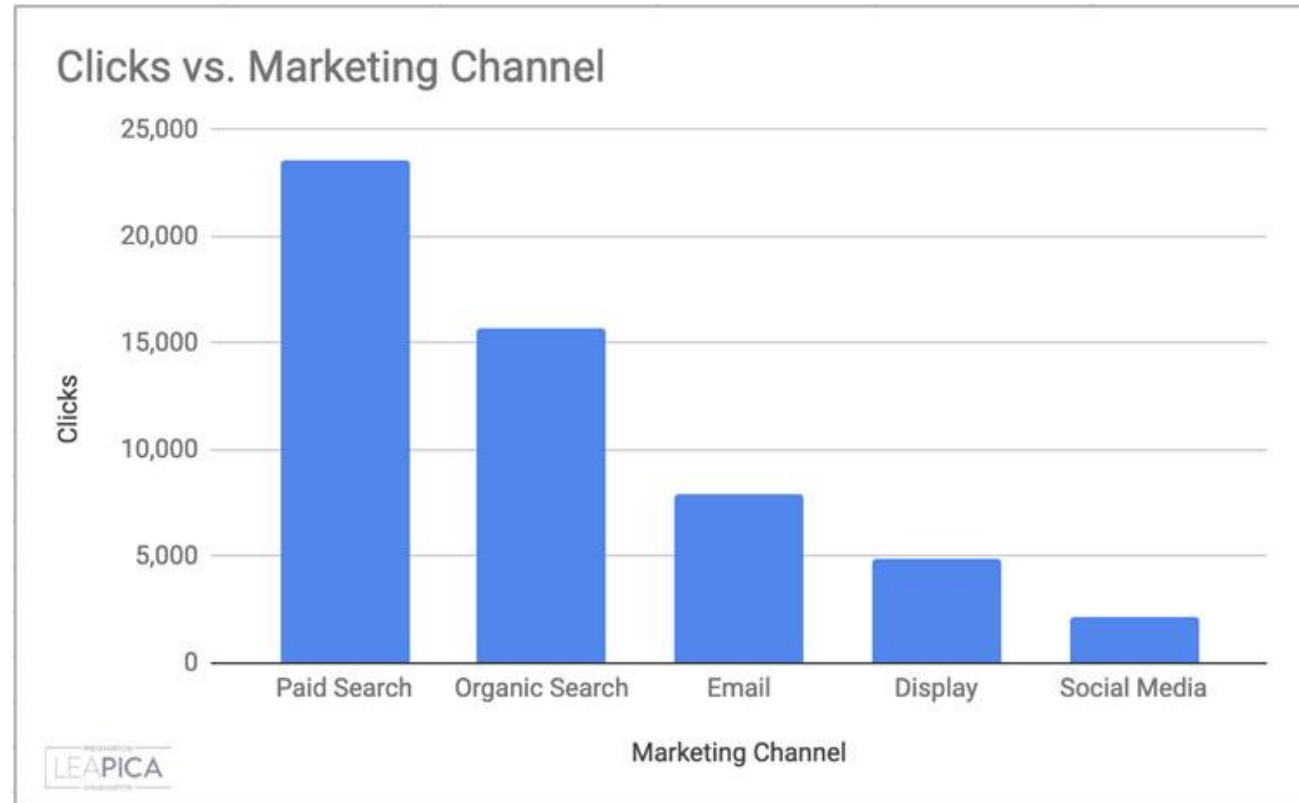


Bar Chart

- A bar graph shows **comparisons** among **discrete categories**.
- **One axis** of the chart **shows** the specific **categories** being compared, and the **other axis** represents a **measured value**.



spain salary by comunidad

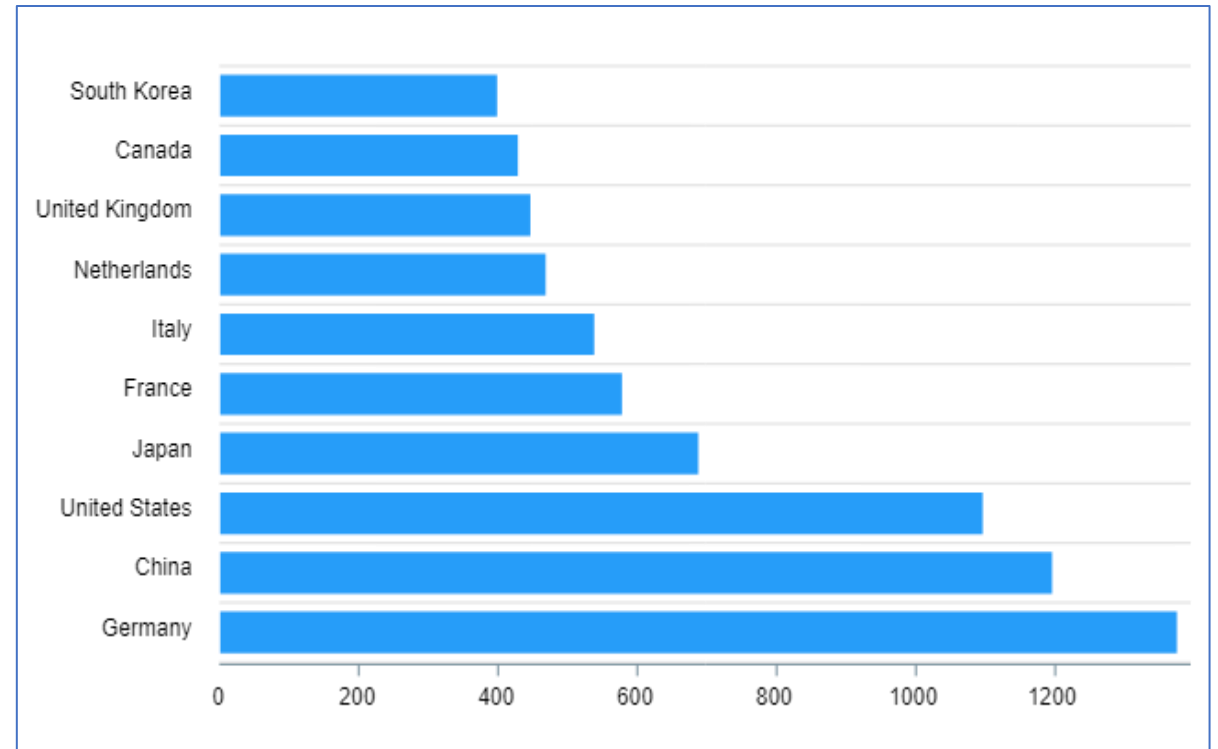


Bar Chart

- Bar graphs/charts provide a **visual presentation** of **categorical data**.
- **Categorical data** is a grouping of data into **discrete groups**, such as months of the year, age group, shoe sizes, and animals.

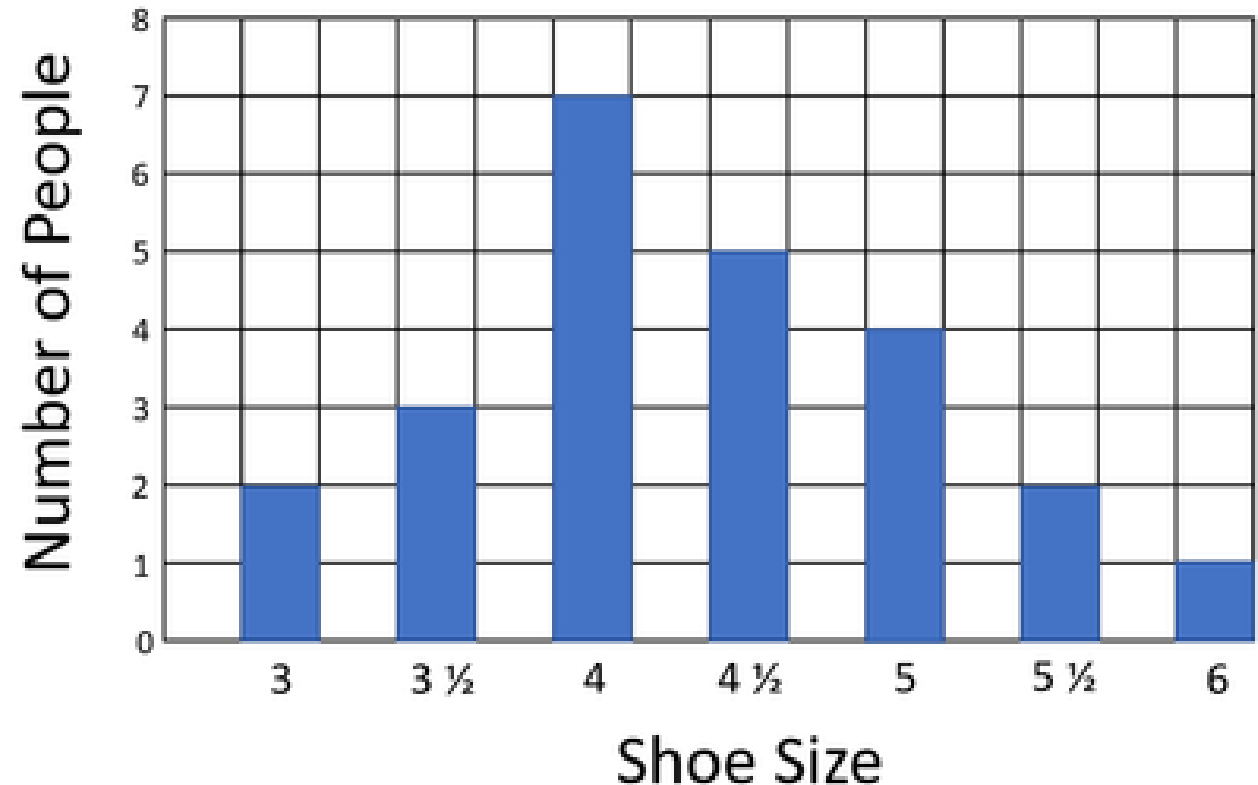


videogame sales



Bar Chart

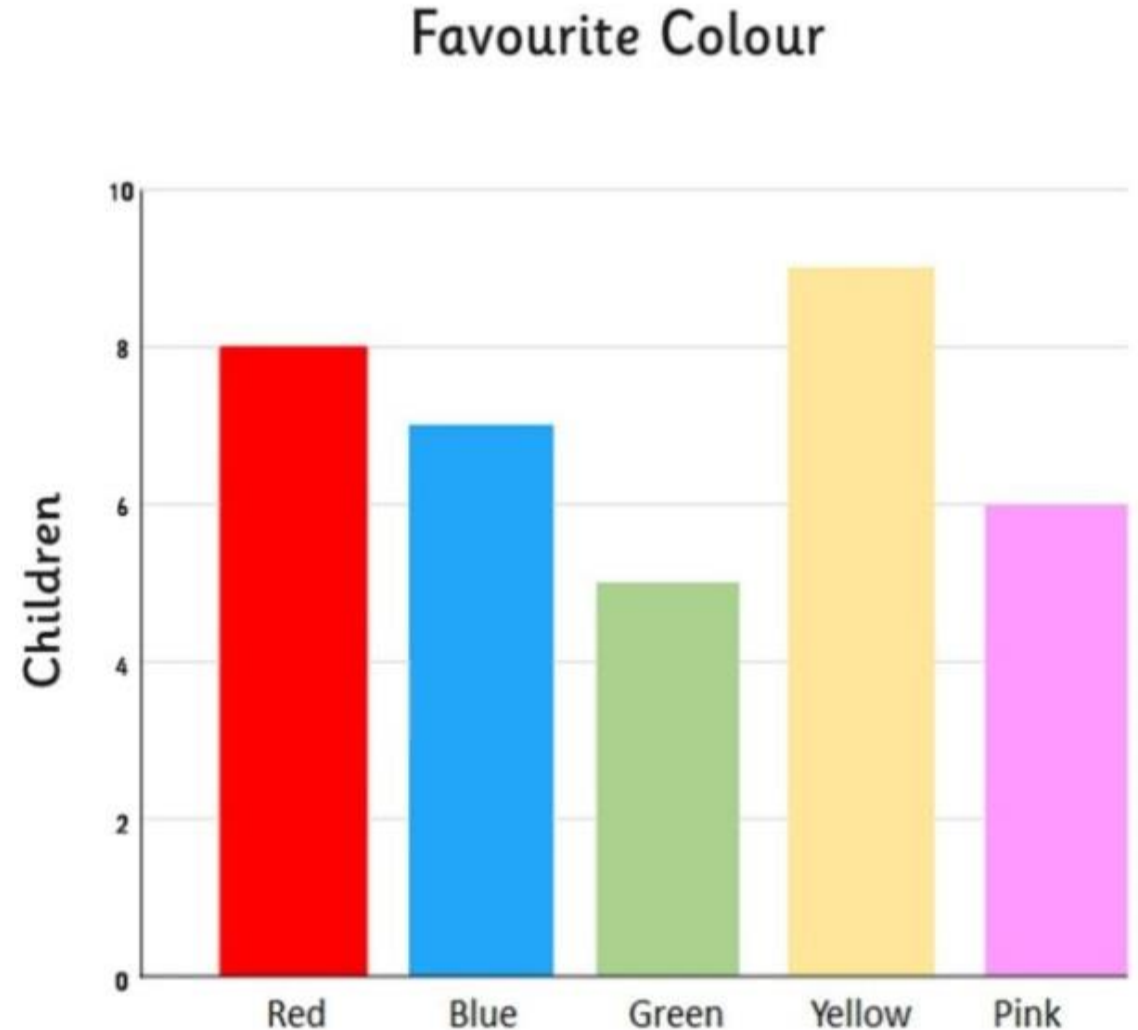
- These **categories** are usually **qualitative**.
- In a column (vertical) bar chart, categories appear along the horizontal axis and the height of the bar corresponds to the value of each category.





Bar/Column Chart - Use

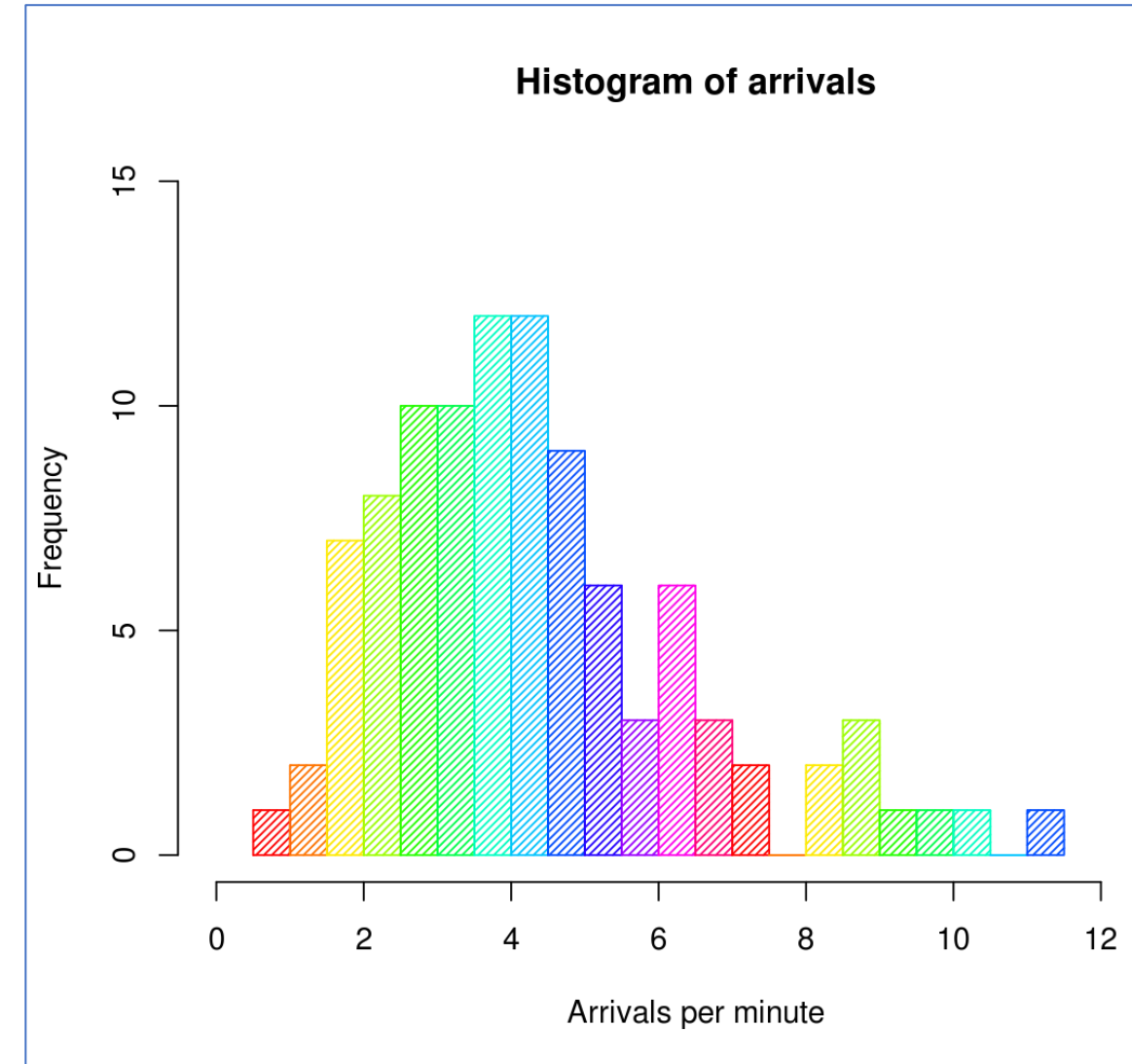
- When you need to **compare** several different categories.
- When you need to show a **side-by-side comparison** of different values.
- When you want to **emphasize** the **difference** between values.
- When you want to **highlight** the **total figures** rather than the **trends**.

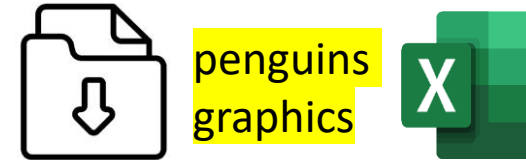




Histograms

- A histogram is an approximate representation of the **distribution of numerical data**.
- To construct a histogram, the first step is to "bin" (or "bucket") the range of values—that is, divide the entire range of values into a series of intervals—and then count how many values fall into each interval.

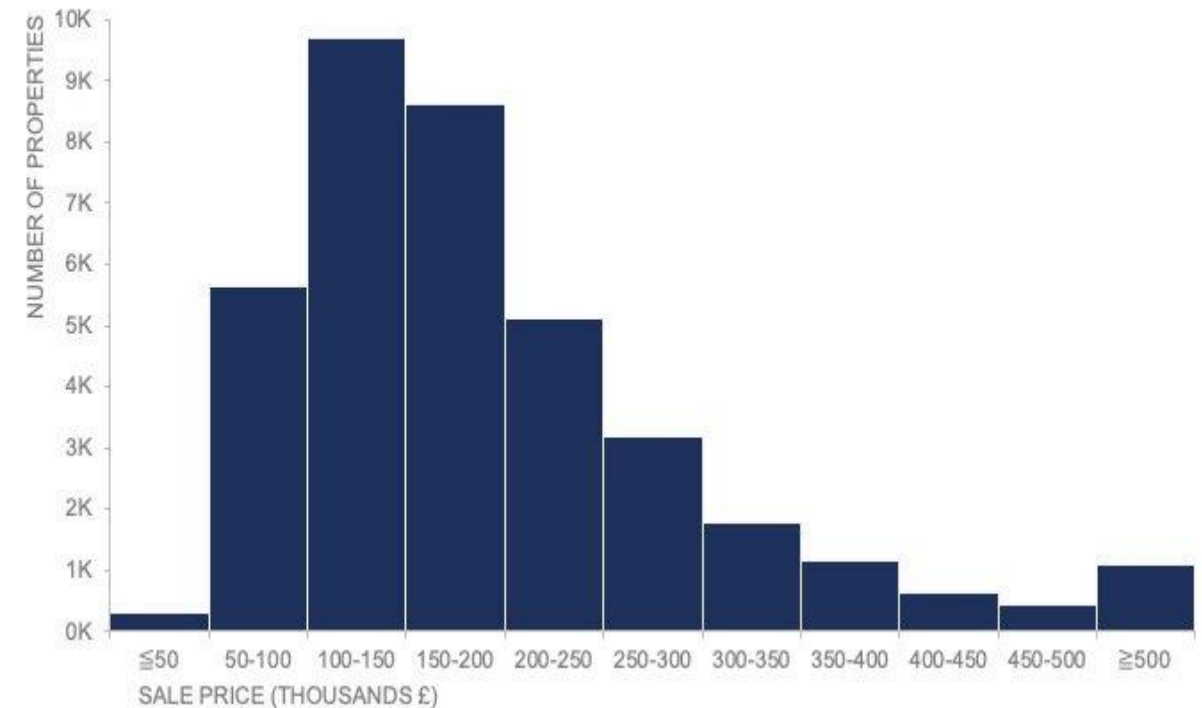




Histograms

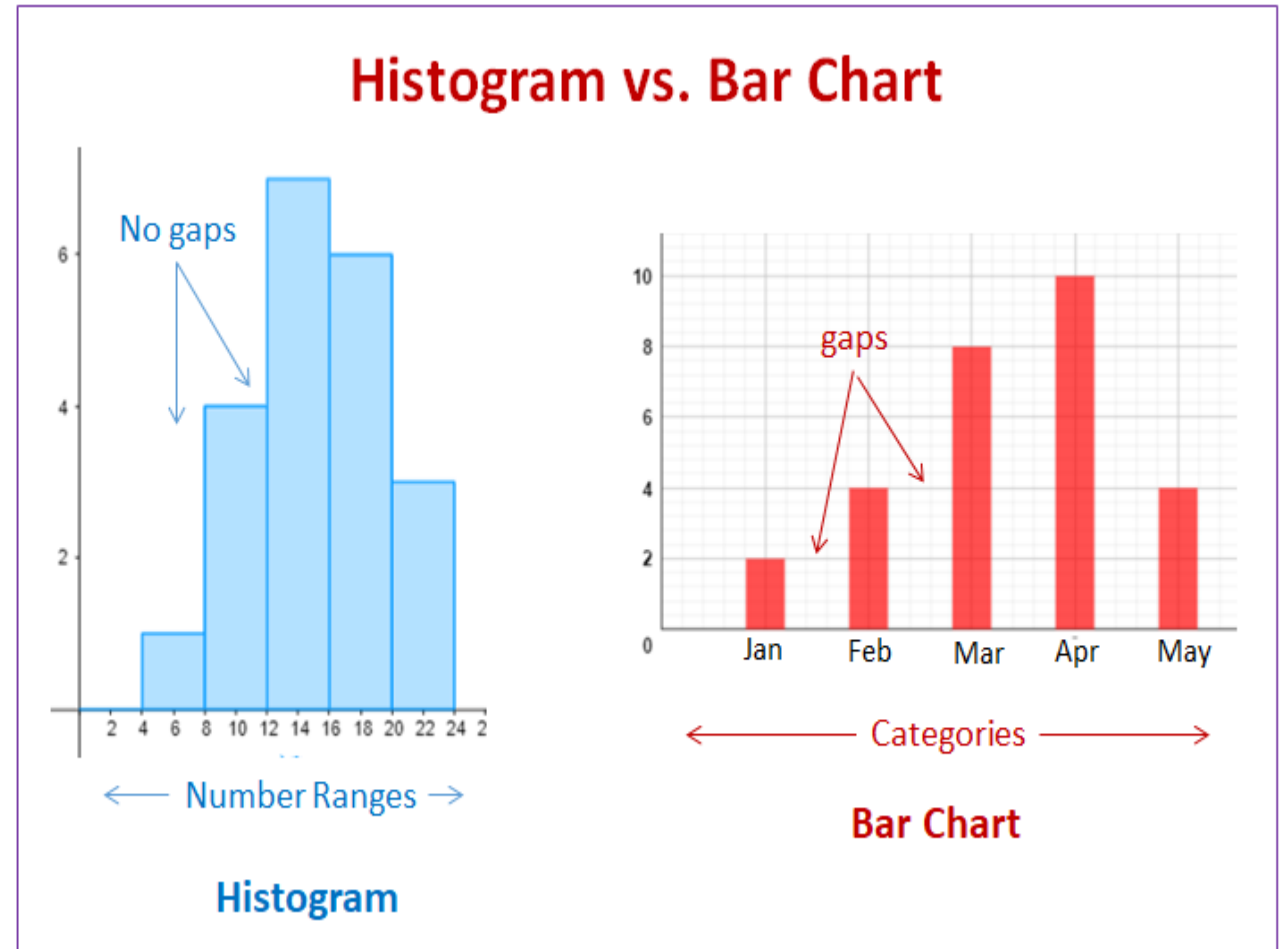
- The bins are usually specified as consecutive, non-overlapping intervals of a variable.
- The bins (intervals) must be adjacent and are often (but not required to be) of equal size.

Distribution of property sales: January 2013 to September 2019



Histograms – Use

- Histograms are sometimes **confused** with bar charts.
- A histogram is used for **continuous** data, where the **bins represent ranges of data**, while a **bar chart is a plot of categorical variables**.
- Some authors recommend that bar charts have **gaps** between the rectangles to clarify the distinction.



Time Series

- A time series is a series of data points graphed in time order.
- Most commonly, a time series is a sequence taken at successive equally spaced points in time.
- For example: the daily closing value of the Dow Jones Industrial Average, or the monthly sales of a company.

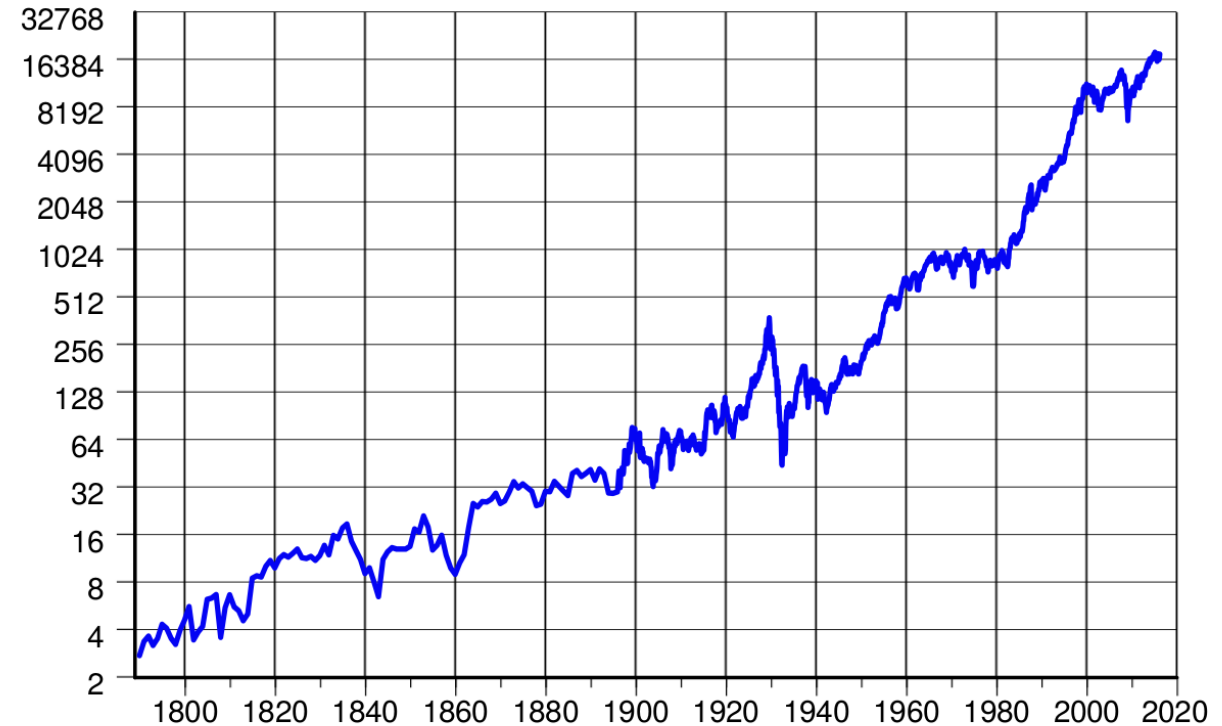


cotizaciones
bolsa



We will use the [STOCKHISTORY](#) Excel function to obtain data for this graph

Dow Jones Industrial Average



Time Series – Use

- When you want to display trends.
- When you want to represent trends for different categories over the same period of time and thus to show comparison.

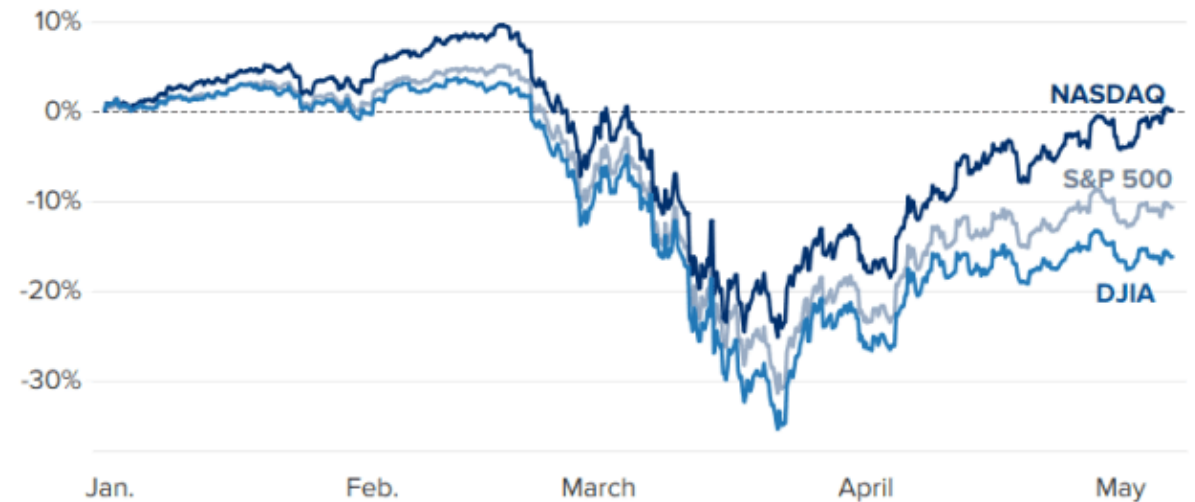


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Nasdaq turns positive for the year

The Nasdaq Composite turned positive for 2020 on Thursday, the first of the three major averages to do so



SOURCE: FactSet. Data as of market close on 5/7/20.

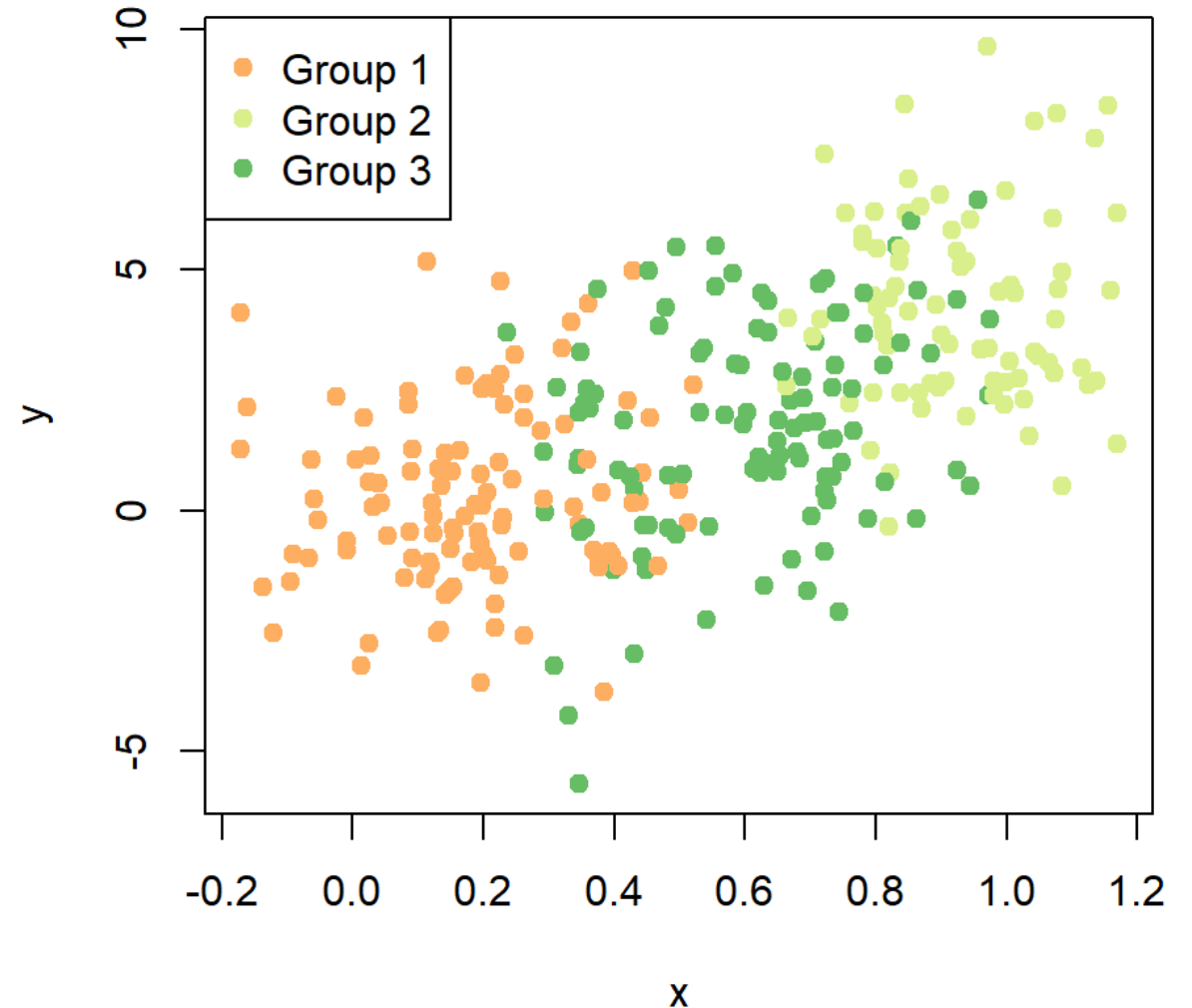


Scatter plot

- A type of plot using Cartesian coordinates to display values for typically two variables for a set of data.
- If the points are coded (color/shape/size), one additional variable can be displayed.



penguins - simpson



Scatter plot – Use

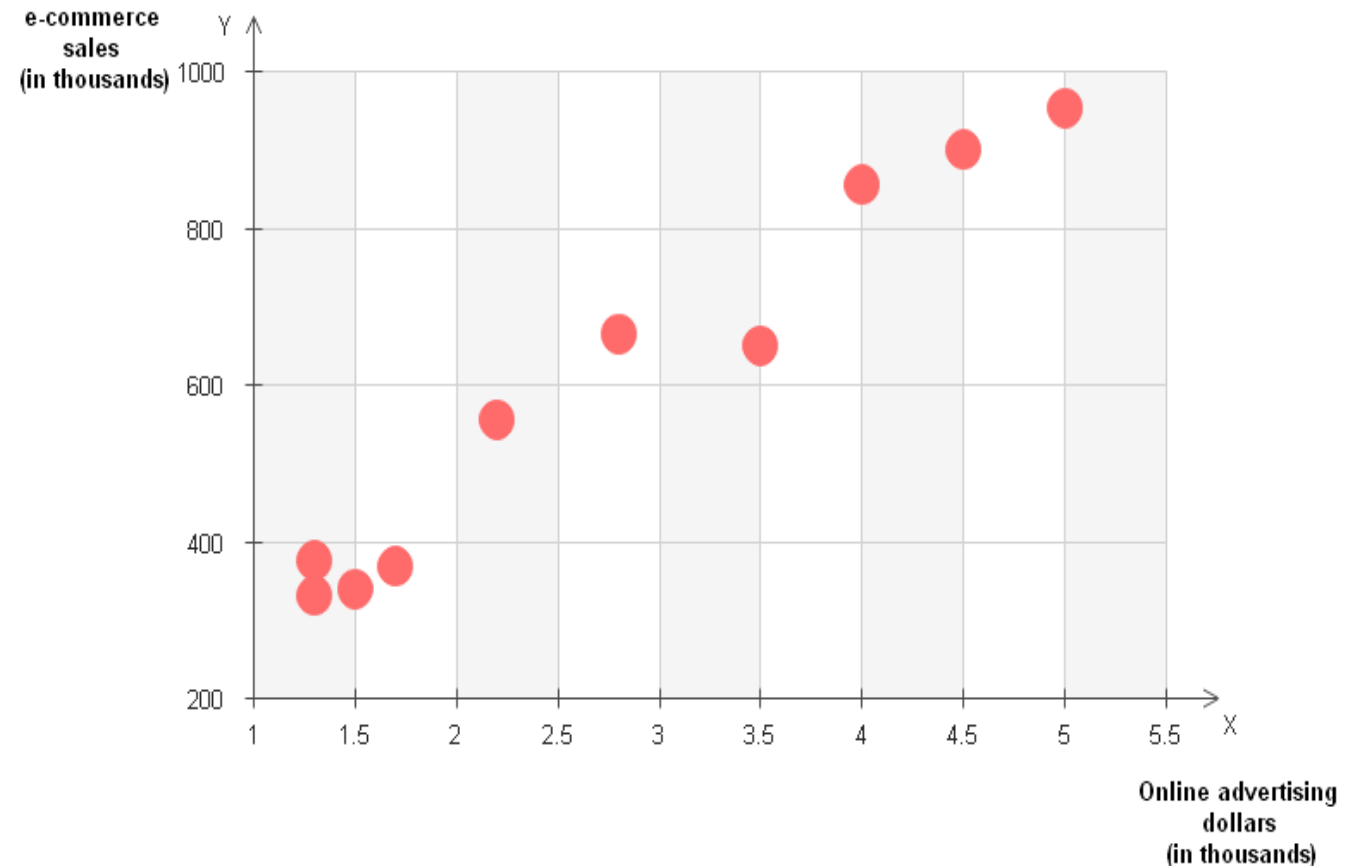
- The data is displayed as a collection of points, each having the value of one variable determining the position on the horizontal axis and the value of the other variable determining the position on the vertical axis.



penguins - simpson



Monthly E-commerce Sales vs Online Advertising Costs



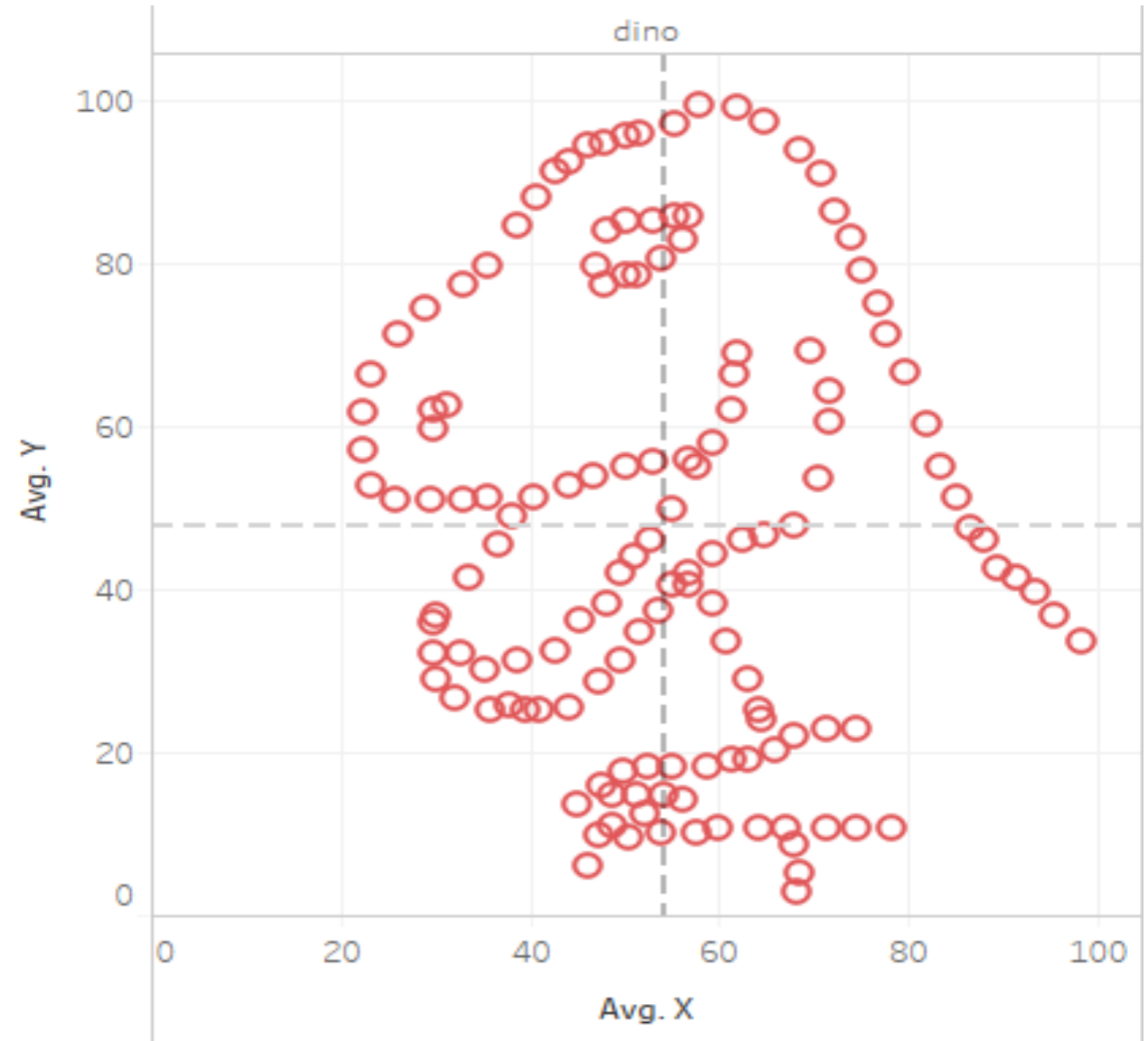


Datasaurus
dozen



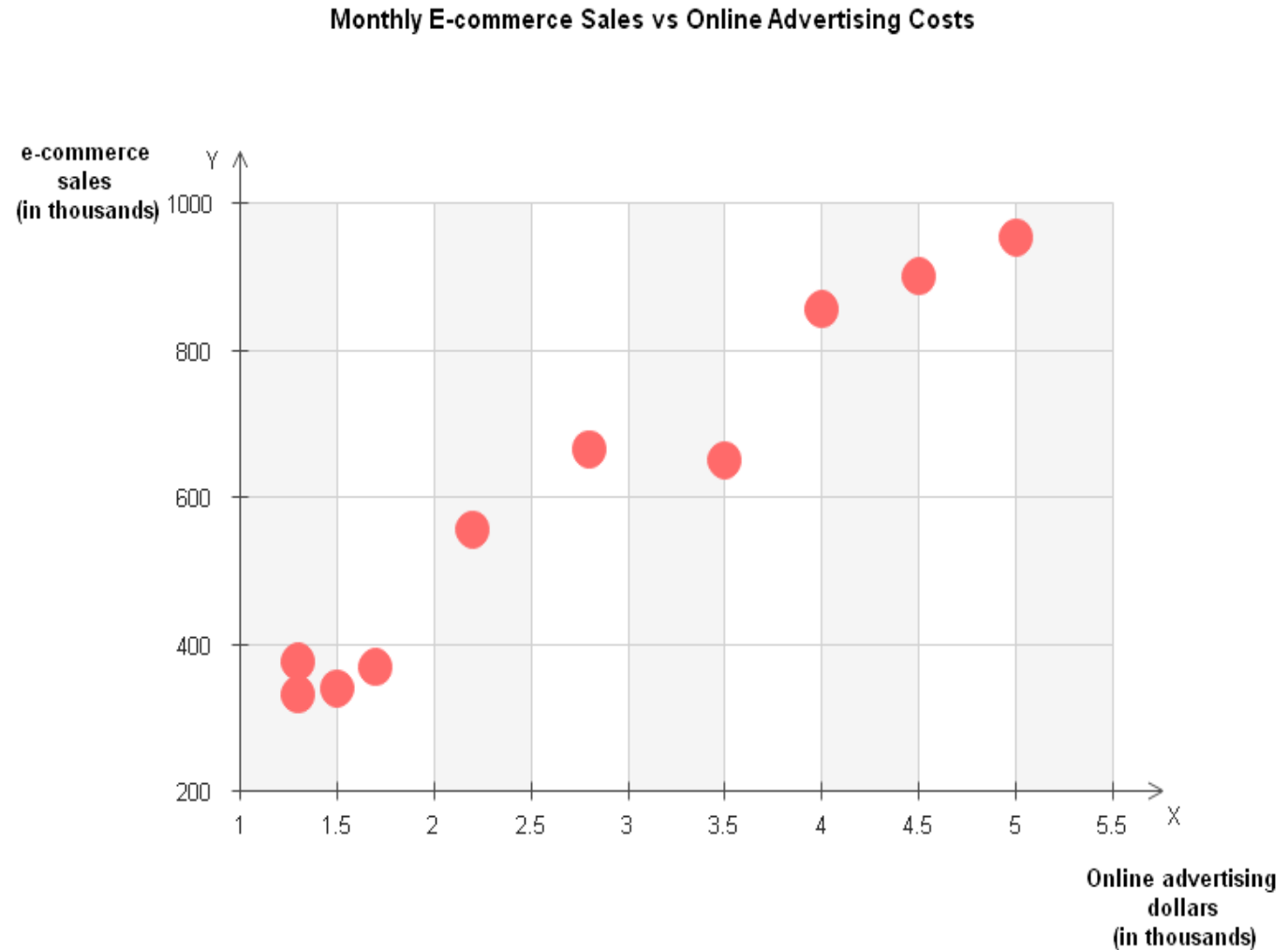
Scatter plot – Use

- All the examples we saw before when talking about the datasets with the same descriptive statistics but different graphs, were graphed using scatter plots.



Scatter plot – Use

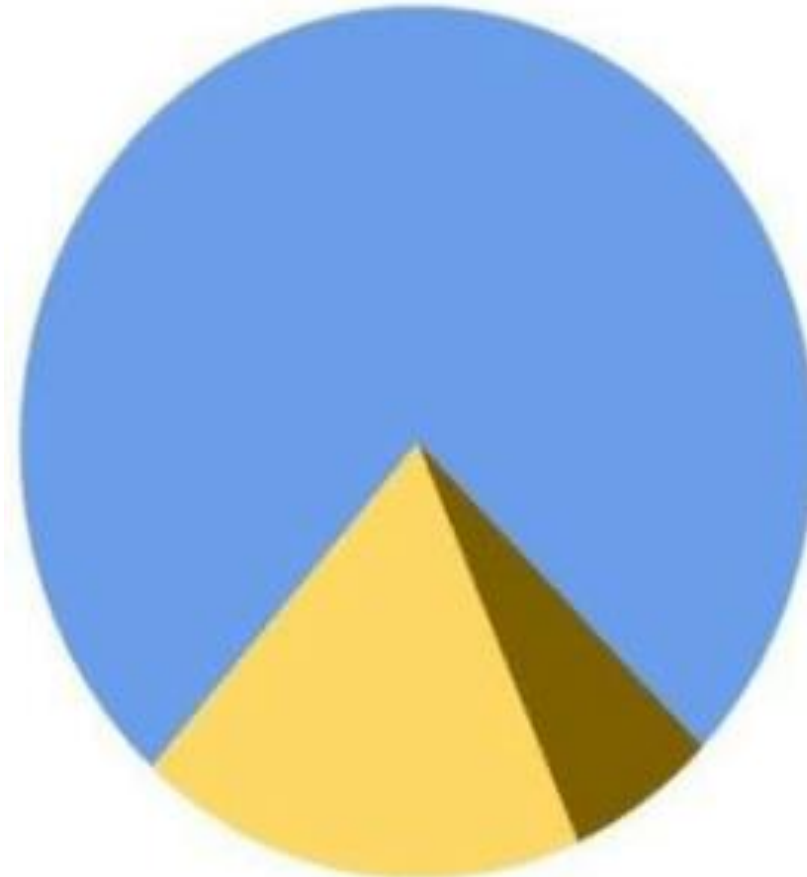
- When you need to observe and show relationships between two numeric variables.





Pie Charts

- A Pie Chart is a circular graph that uses “pie slices” to display relative sizes of data.



Sky



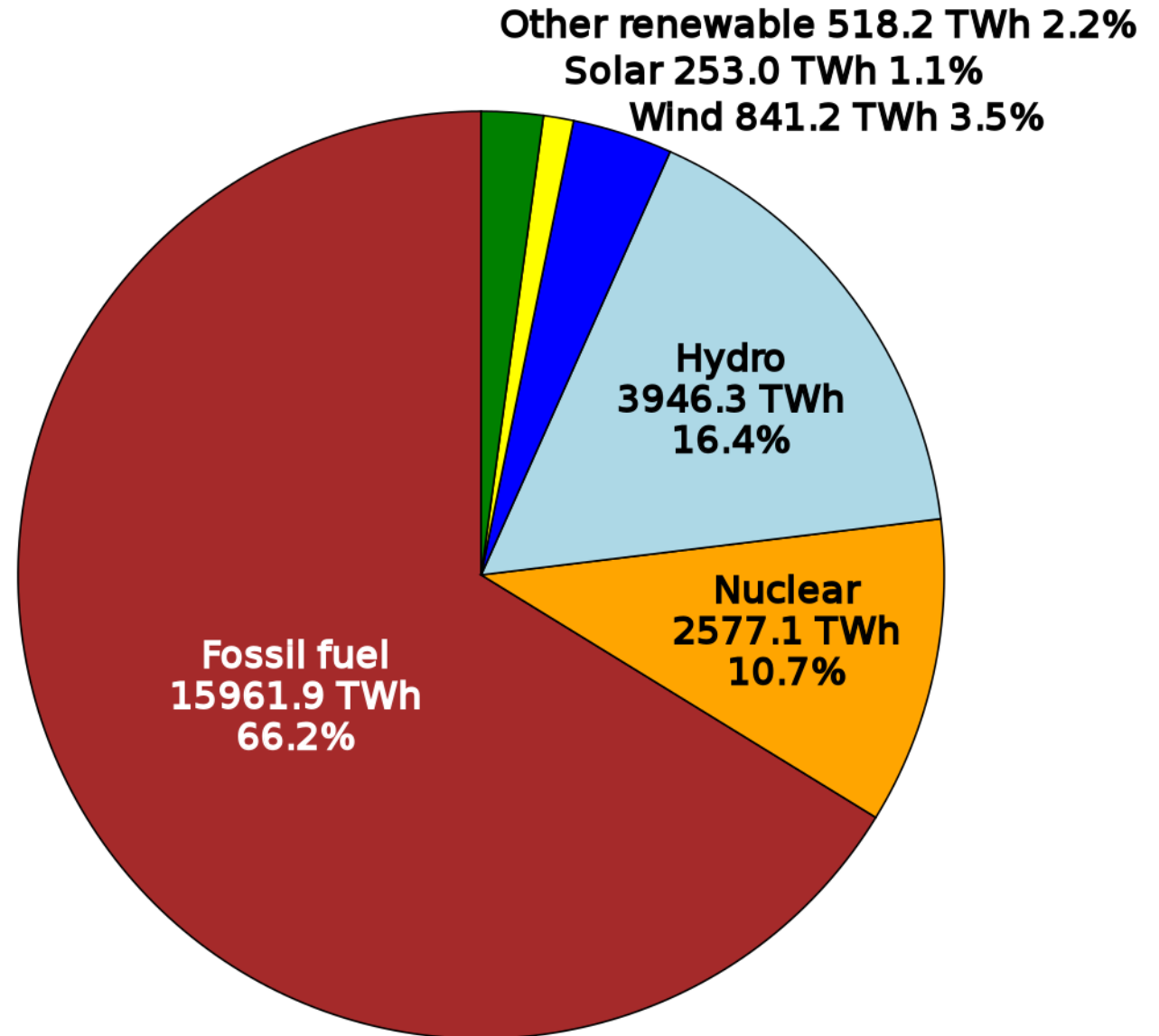
Sunny side of pyramid



Shady side of pyramid

Pie Charts - Use

- When you want to represent the share each value has of the whole.
- When you want to show how a group is broken down into smaller pieces.



World Electricity Generation by Source (2015)

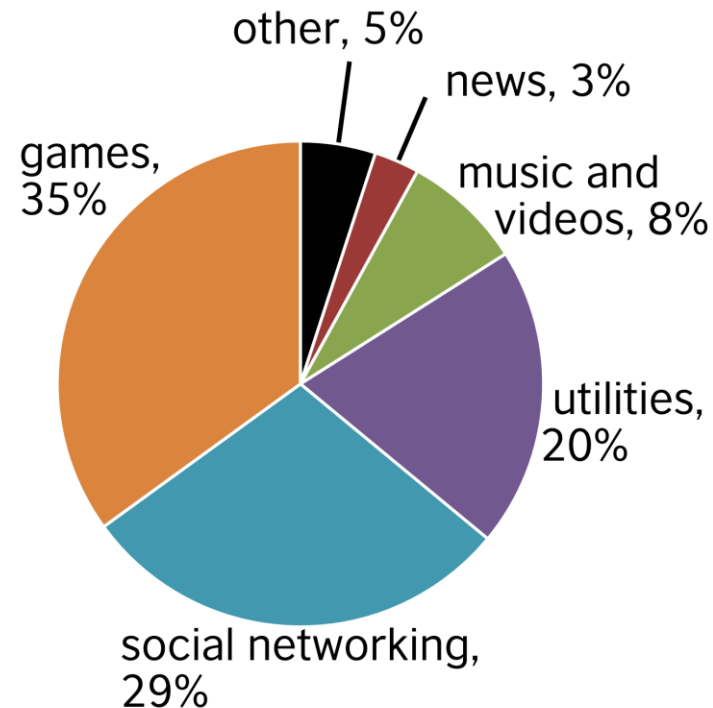


Pie Charts

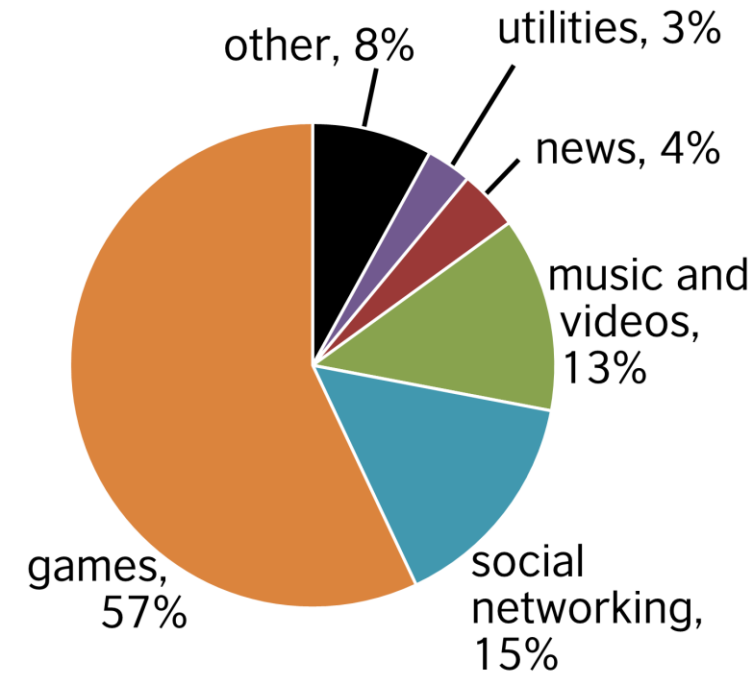
- It is difficult to compare different sections of a given pie chart, or to compare data across different pie charts.

Time spent on smartphones and tablets, by category

Smartphones



Tablets

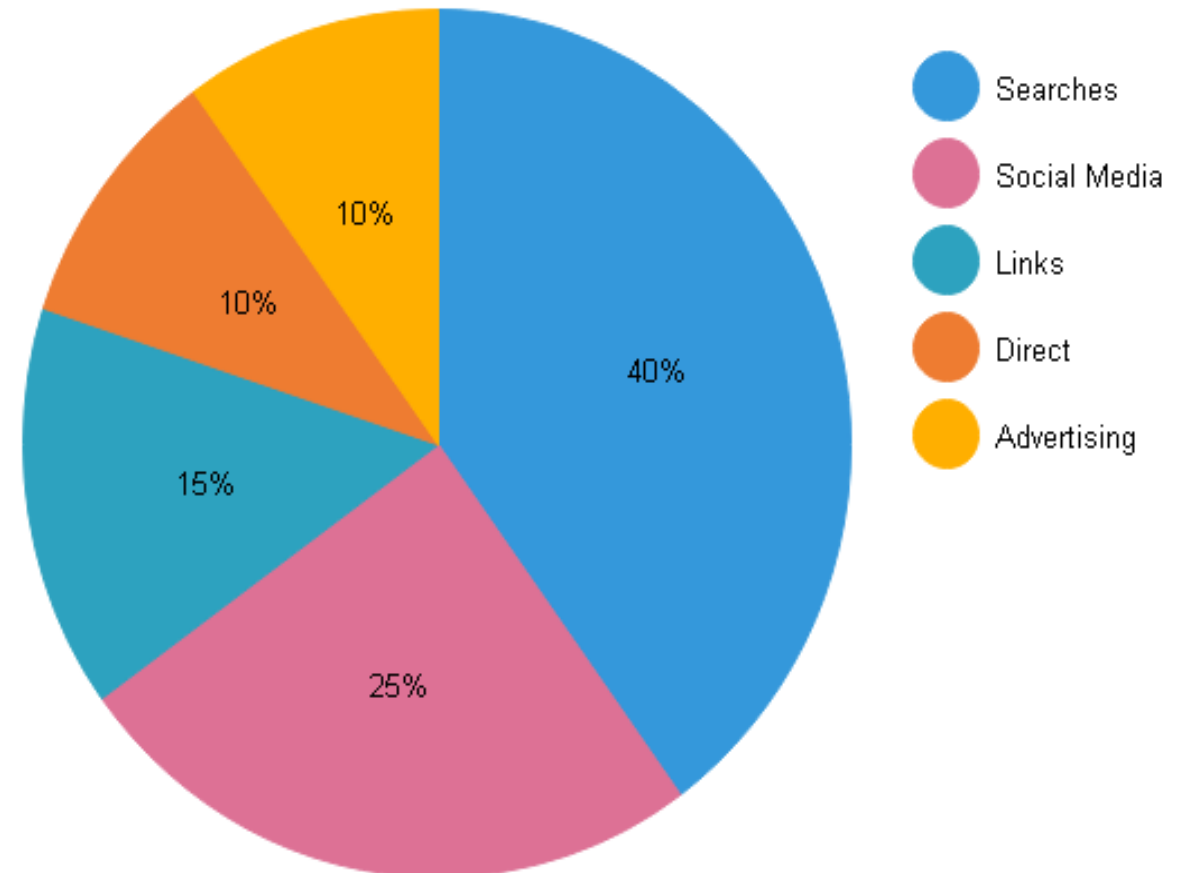




Pie Charts – When to use them

- A pie chart is a good choice for visualizing percentages because it shows each element as part of a whole.
- The entire pie represents 100 percent of a whole. The pie slices represent portions of the whole.

Visitors By Web Traffic Sources

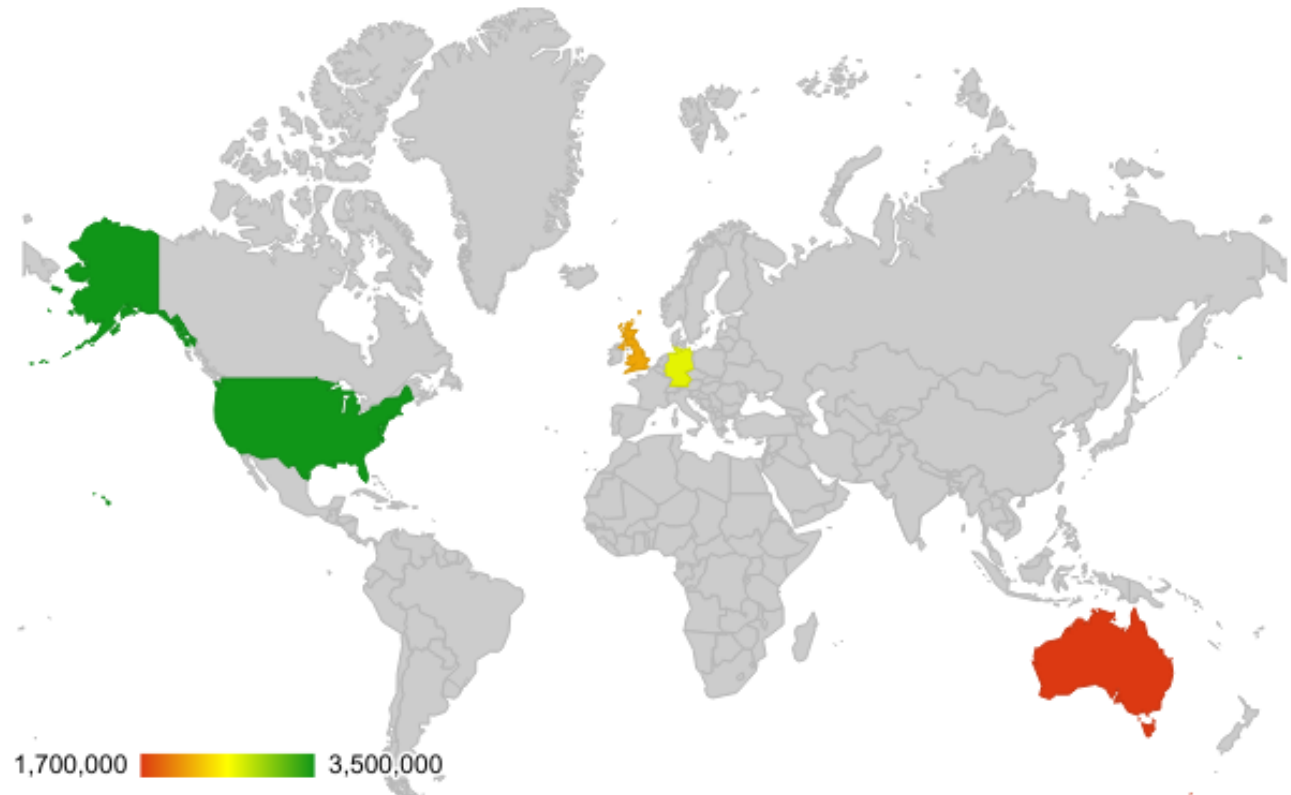




Geospatial

- Using actual maps to represent the information from countries or other geographical divisions.

Region	Sales
United States	\$3,500,000
England	\$2,200,000
Australia	\$1,700,000
Germany	\$2,700,000



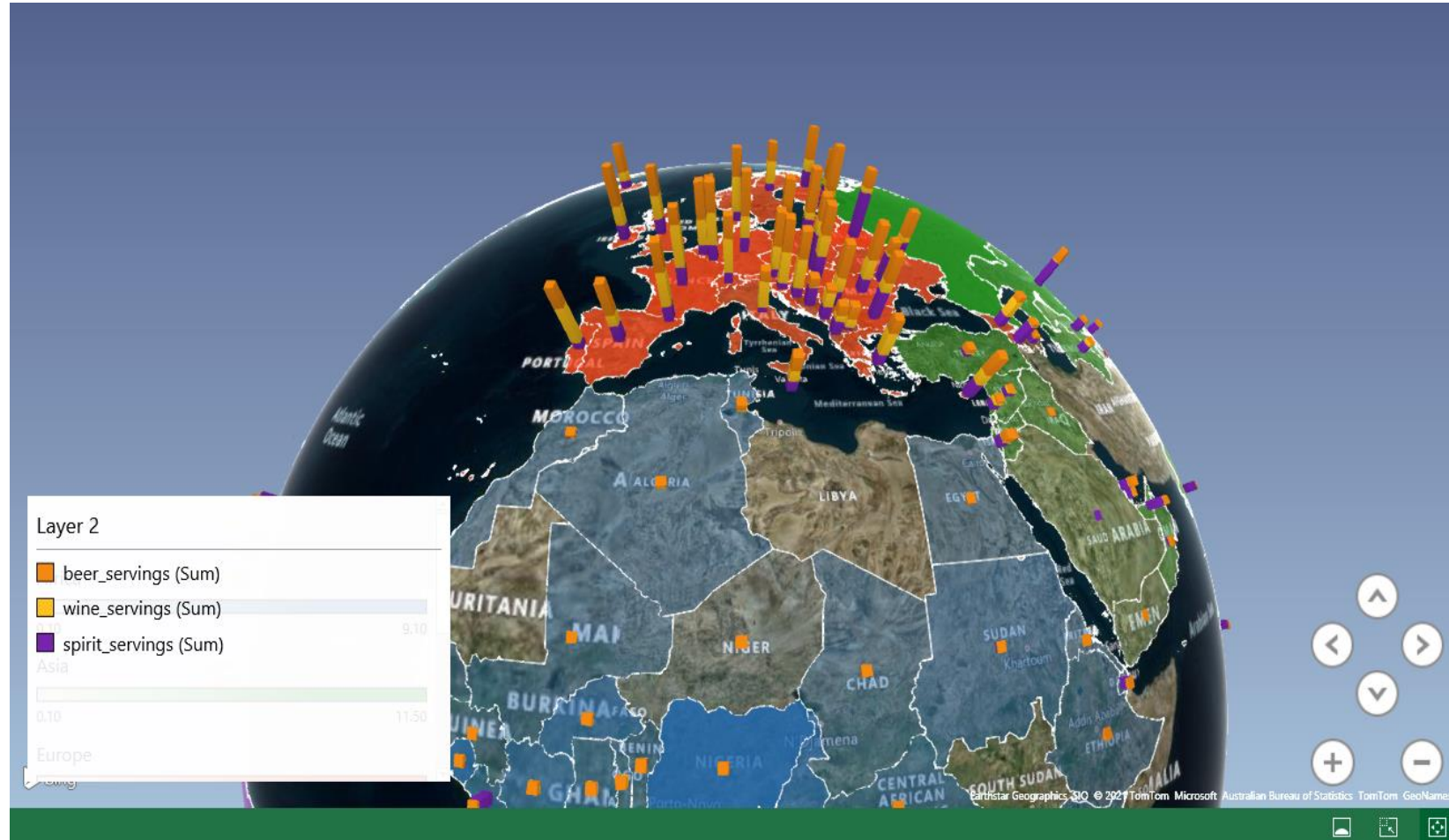


drinks by
country



Geospatial- 3D

- Using 3D maps to represent the information from countries or other geographical divisions.



<https://www.youtube.com/watch?v=P--qP4mfxEg>



Takeaways for the Topic –Descriptive Analytics

- EDA is exploratory in name and nature.
- Descriptive Analytics can help with EDA.
- Descriptive Statistics help us understand and compare the underlying distribution of variables.
- Graphs and charts contribute to our understanding of datasets.
- Data needs to be thoroughly explored before it can be understood and used for analysis.
- Different charts need to be used for different types of data.